Learning Distributed Algorithms from Algorithmic Teachers

Research collaboration (author order pending)

23 October 2026

Unpublished integrated research draft, 23 October 2026. Author list and order remain to be agreed. Historical contributions are attributed by Git provenance.

Abstract

Learning a distributed algorithm can mean predicting its actions, reproducing its transitions, solving its task, matching its running time, or generating the same probability law over solutions. These objectives need not coincide. We investigate distributed graph coloring through seven explicitly scoped algorithmic teachers and shared neural policies. A historical five-seed architecture study finds that normalized node/color networks, DeepSets palette scorers and local palette Transformers each complete all 1,520 held-out evaluations, while aggregation and representation controls expose failures. A complementary seven-teacher study shows that almost-perfect local action accuracy can coexist with substantial autonomous failure. We integrate the three palette architectures with every native teacher, add small and no-communication controls, and train 70 balanced short-budget replicas. We separately audit teacher safety and termination, compare independent scalar specifications on 103 small graph representatives, and distinguish supplied arithmetic and schedules from learned decisions. Statistical reconciliation aligns the compared output laws and shows agreement between two procedures on six identical archival banks. Fresh known-law controls reveal a large power difference for localized alternatives. Our evidence supports a scoped separation between functional generalization and fidelity to a teacher's complete solution law. Missing historical collaborator weights prevent a complete numerical replay of that cohort; no conclusion imputes those artifacts or turns a nonrejection into equality.

A separate controlled-supervision study trains 735 fresh policies across seven teachers, three architectures, seven label budgets and five training seeds. Its strict benchmark certifies 12 of 21 teacher/architecture packages only when all five policies complete 1,000 final executions each without failure. Explicit information regimes, nonmonotone per-seed curves and fixed-label optimization controls separate this finite empirical certification from universal correctness and theoretical sample-complexity claims.

A counterfactual extension adds admissible phase-specific oracles and fixed-label optimization controls. It closes all nine previously pending teacher/core packages on fresh five-policy, 1000-execution batteries while preserving the twelve historical certifications. Concordant controls also close those cells; counterfactual continuation fidelity and required optimizer effort differ. Complete-output coupling, information ablations and larger-graph transfer delimit what functional success establishes.

1 Introduction

A distributed coloring algorithm is an executable local rule embedded in a communication model. A neural learner trained from its trajectories must represent that rule, survive the states induced by its own decisions, and remain meaningful when the number of neighbors or colors changes. A proper coloring at the end of an execution is one useful outcome, but it leaves several questions unanswered. Did the network make the same decisions? Did it use the same information? Did it preserve the teacher's progress guarantee? Did it sample the same coloring distribution?

These questions become sharper when teachers differ. Random trials use private randomness to resolve conflicts; priority rules exploit distinct identifiers; color-wise preferences compare several possible colors; auxiliary color reduction uses finite-field representations; speculative programs repair temporary conflicts; and sequential greedy coloring assumes a supplied order. A common neural interface can compare their local decisions only if those differences remain explicit. Equal parameter counts do not compensate for different supplied information or computational schedules.

We study five notions of learning: **action fidelity**, agreement on common teacher states and private inputs; **transition fidelity**, agreement of updated states; **functional validity**, producing a proper complete coloring; **complexity fidelity**, preserving an appropriate teacher-specific cost; and **distributional fidelity**, agreement of the probability laws of complete solutions on a fixed labeled graph. The central question is how these notions relate across teacher, architecture and evaluation regime.

The project combines two complementary lines. The degree-generalization studies emphasize variable palettes, neighborhood aggregation, conditional safety/progress and full labeled output distributions. The native-teacher studies emphasize distinct algorithms, supplied phase interfaces, local imitation and autonomous size transfer. Their results are not pooled as if they used one protocol. Instead we reconstruct the definitions, preserve their separate cohorts and add a balanced integration experiment.

We add a second central axis: how much supervision each architecture needs to achieve perfect functional performance within a defined benchmark. An acquired node/action-component label is counted separately from a trajectory, round, data replay or optimizer update. Information compatibility precedes the comparison. The primary empirical gate requires all five independently trained policies to complete every final execution without a single failure; the observation and optimization budgets remain separate experimental factors.

Our strongest conclusion is deliberately qualified: some learned local policies generalize functionally while differing from their teacher in observable full-output laws. This is not a claim that every successful model must differ, or that every difference can be detected at our sample sizes. Symmetry can force equality on particular graphs, and a valid coupling can certify approximate agreement on a fixed graph. A coherent account must accommodate all of these outcomes.

1.1 Contributions and their status

First, we specify a teacher/architecture interface preserving seven native action alphabets and their supplied information. E1, DeepSets and local palette attention are represented for each teacher, with a controlled short-budget experiment and two additional baselines. This is a framework and empirical study, not seven newly invented distributed algorithms.

Second, we provide conditional proofs for the coded exact teacher rules, an independent scalar tran-

sition audit and explicit limits of finite testing. A repaired input-domain guard has a saved failing regression. We separate correctness of a mathematical teacher from correctness of an arbitrary trained numerical policy.

Third, we reconcile the statistical objects before comparing findings. Matched-bank replays preserve original statistics; fresh known-law controls quantify a finite-sample sensitivity difference. Existing factorial data add observable-event bounds and size/degree interactions without asserting a monotone law for total variation.

Fourth, the repository supplies an evidence ledger, complete scientific audit, frozen checkpoint manifests, historical negative results, executable reproduction commands and remotely verified recovery artifacts. Historical collaborator checkpoint-level reproduction remains explicitly incomplete.

2 Related work

Neural algorithmic reasoning studies learned execution of algorithmic computation [Veličković and Blundell, 2021]. CLRS supplies a common benchmark spanning classical algorithm families [Veličković et al., 2022]. Our question focuses on repeated local decisions in specified distributed interpreters, including randomness and stopping semantics, and evaluates a solution law as well as task success. We do not claim that trajectory imitation or algorithmic generalization is new.

Distributed locality restricts the information available within a bounded communication radius [Linial, 1992]. Connections between local algorithms and GNN expressiveness clarify the importance of inputs, depth and aggregation [Sato et al., 2019, Loukas, 2020]. Those results do not imply that a particular optimized network generalizes correctly. Our controlled observations and conditional proofs concern explicit architectures and teachers, rather than a universal GNN expressivity theorem.

Deep Sets gives a principled shared-function/pooling construction for sets [Zaheer et al., 2017], while attention supplies content-dependent interaction [Vaswani et al., 2017]. Here both act over each node's palette; graph communication remains a neighborhood operation. Our positional palette features retain color-name information, so these models must not be described as fully color-permutation invariant.

Imitation learning encounters a state-distribution shift when learner actions affect later observations [Ross et al., 2011]. We use teacher trajectories and counterfactual COMMIT labels, and evaluate autonomous rollouts. This is not equivalent to on-policy dataset aggregation. Work on asynchronous GNN inference also warns that synchronized behavior need not survive changed execution semantics [Solodova et al., 2025]; this paper does not test asynchronous correctness.

The teachers draw on randomized coloring [Luby, 1993, Johansson, 1999], priority coloring [Jones and Plassmann, 1993], polynomial and iterative reduction [Linial, 1992, Kuhn and Wattenhofer, 2006, Barenboim and Elkin, 2008], color-wise preference trials [Schneider and Wattenhofer, 2010] and speculative repair [Gebremedhin and Manne, 2000]. We state the implemented specialization rather than transferring the full complexity guarantee of each cited paper.

Kernel two-sample tests compare probability laws through MMD [Gretton et al., 2012]. Characteristicness is an identifiability property, not a finite-sample power guarantee. Random-permutation calibration and familywise correction require their own assumptions [Hemerik and Goeman, 2018, Holm, 1979]. Our honest witness omnibus is a specified empirical instrument; no general minimax

optimality or universal dominance is claimed. This is a scoped primary-source review, checked on 23 October 2026, rather than an exhaustive novelty determination.

3 Formal model and notions of fidelity

Let $G = (V, E)$ be a fixed finite simple undirected labeled graph, $n = |V|$, and let C_v be the allowed colors of vertex v. A state contains a partial coloring $q_v \in C_v \cup \{-1\}$ and any explicitly declared auxiliary variables. A proper partial state has no equal fixed colors across an edge. For permanent-color trial teachers,

$$A_v(q) = C_v \setminus \{q_u : u \in N(v), q_u \geq 0\}.$$

In the local-palette studies, $C_v = \{0, \dots, d(v)\}$. Native degree-three comparisons use a common $C_v = \{0, 1, 2, 3\}$ even at vertices of lower degree. The two settings coincide on cubic graphs and generally differ on irregular ones. No common maximum-degree input is supplied to the local-palette E1 policy. Some native adapters explicitly use the known bound 3.

A protocol step maps a current state and private input to participation B, candidate C, confirmation J, an auxiliary choice, or a repair mark, according to its native phase. A permanent fixation is applied only at a pending, permitted participating node with a candidate and positive COMMIT. Palette range, pending-node gating, native phases and schedules are supplied mechanisms; availability and conflict decisions are learned unless explicitly identified as an exact-teacher baseline.

For a frozen teacher T and policy M, action fidelity evaluates their decisions on identical input states and explicit random arrays. Transition fidelity also evaluates the resulting states. Functional success is the event that all vertices receive an allowed color and every edge has different endpoint colors. These three predicates need not coincide. A policy may produce another valid solution after its first action disagreement; an inaccurate decision may instead propagate to a later conflict.

For cutoff R, write S_R for proper completion and $P^G_{T,R} = \mathcal{L}(X_G \mid S_R)$, where $X_G = (c_1, \dots, c_n)$. Vertex and color names remain fixed. A joint terminal law instead observes colors, partial outcomes and status at R. A quotient law canonicalizes color names. These are different estimands. If $P(S_R) = 0$, the conditional solution law is undefined and cannot be estimated by resampling until success.

Total variation is $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. A statistically significant fixed-event difference therefore gives a TV lower bound. For any valid coupling (X, Y), $\mathrm{TV}(P, Q) \leq P(X \neq Y)$ instead gives an upper bound. Large disagreement under a poor coupling supplies no lower bound. In particular, shared-randomness action fidelity and solution-law fidelity are distinct questions.

4 Algorithmic teachers

Table 1. Implemented teachers and computational scope. All native comparisons below have degree$\leq$ 3 and global K4. Their exact references and proof assumptions are expanded in AppendixA and the teacher-audit document.

Teacher	Local decision	Supplied information and cost convention
LUBY	Bernoulli 1/2 participation; random available rank; reject equal neighboring proposals	Two exchanges per trial; no IDs needed
JOHANSSON	Always-active random available rank; reject equal neighboring proposals	Common K4; two exchanges per trial
JP	Pending strict ID maxima take first free color	Fixed distinct IDs; synchronous native rounds, capped 128
COLOR_TRIALS	Strict discrete preference winner per color; keep lowest winning color	Preferences 0–3, availability exchange; repeated subroutine
BE_BASE_D3	Linial field-point reductions then KW recoloring	Phase clock, polynomial evaluation and receiver-relative equalities supplied
SPECULATIVE_GM	Concurrent block assignment, same-slot detection, serial repair	Four balanced blocks and writable schedule supplied; intermediate conflicts allowed
GREEDY	One permitted vertex takes first free color	Sequential ID-order schedule supplied

The word LUBY denotes the explicit coin-participation random-trial rule. It does not mean that this repository implements every derandomization or PRAM component of Luby's work. JOHANSSON uses a larger common initial palette at lower-degree vertices; the original analysis permits larger initial lists, but the output law differs from the smallest local-list initialization. Repeated COLOR_TRIALS substitutes a fixed global degree bound for its neighborhood bound and omits RankingTrials. BE_BASE_D3 is the degree-three base construction with two explicit Linial steps, not the high-degree recursive algorithm. GM and greedy retain schedules that would require a separate distributed implementation if no scheduler were supplied.

The four parallel trial/priority interpreters stop at 128 native steps. BE uses its auxiliary phases; GM and greedy run their supplied schedules. The historical canonical E1 behavior experiment uses cutoff 256 and continues after a collision. We retain these differences rather than calling all reported native steps a common LOCAL running time.

5 Neural policies

5.1 E1 NORMALIZED

For each vertex/color pair define pending status u_v and ownership $o_{vc} = \mathbf{1}[q_v = c]$. Shared learned functions send a nonnegative message and form a local state,

$$m^1_{vc} = \phi_1(u_v, o_{vc}), \qquad h_{vc} = \psi_1(u_v, \max_{w \in N(v)} m^1_{wc}).$$

Availability is learned as $\alpha_{vc} = \sigma(a(h_{vc}))$. The ordered prefix $S_{vc} = \sum_{j<c} \alpha_{vj}$ and

$D_v = \max(\sum_j \alpha_{vj}, .5)$ give normalized endpoints $L_{vc} = S_{vc}/D_v$, $U_{vc} = (S_{vc} + \alpha_{vc})/D_v$. A shared scorer reads h, α, L, U, r_v^c; the candidate is the maximum scoring palette slot. Occupied colors are not removed by an exact mask.

The second exchange represents pending/ownership/participation/offer bits. A learned COMMIT head reads the candidate slot after neighbor MAX. Its prediction is applied without an exact safety veto. Canonical E1 has 16,035 parameters with hidden/message dimensions 64/32. This count is independent of palette size, but messages, memory and computation grow with the palette; each directed edge carries 32K floats per message stage in the direct implementation.

5.2 DeepSets and local palette attention

Both variants retain learned neighborhood messages, availability and COMMIT. DeepSets replaces ordered-prefix scoring with shared per-slot features and a mean of learned palette features. LocalPaletteTransformer projects slots to dimension 16 and uses two-head self-attention with a 32-dimensional feedforward layer and zero dropout. Its attention is within one vertex's palette, not between arbitrary graph vertices.

Their slot inputs include learned availability, normalized color position, inverse palette size and private randomness. Prefix-free is therefore not position-free. All three models are vertex-equivariant in the ideal shared-rule, invariant-aggregation model, while ordered prefixes and positions generally break color-name permutation symmetry. Floating-point backend probes support finite implementations but do not prove universal numerical equivariance.

5.3 Native teacher adapters and baselines

The common native interface preserves Pedro's two-hop rooted representation and exact action alphabets. LUBY/Johansson/greedy use the unextended core. JP, ColorTrials, BE and GM retain supplied native context residuals. The E1 version adds 3,361 parameters in those cases; analogous context is retained for the other palette architectures. ColorTrials selects its final candidate after the second message. BE-Linial predicts a valid field point and computes the polynomial value through declared local arithmetic. The DeepSets and Transformer masks use the actual field/palette size, including mixed field 7/13 examples.

Two new controls use the same supervision and native action interfaces. The **small** model reduces hidden/message dimensions to 8/4 inside the normalized family. The **no-communication** control erases received neighbor states, relations and masks while retaining own state and supplied native permissions. It is an information ablation with the same large core, not a claim of parameter minimality. Exact teachers are zero-trainable-parameter task baselines. The earlier direct-action 36,719-parameter GNN remains a historical comparator; it uses different representation/loss choices.

No architecture is called minimal merely because it has fewer parameters. We distinguish a mathematical information limitation, sufficiency within a tested architecture family, and finite empirical success.

6 Training and experimental design

6.1 Three distinct training cohorts

The historical mechanism study trains five seeds 41000–41004 per primary architecture on 800 cubic N40 graphs, three teacher trajectories,120 epochs, Adam at .001 and batch 128 round examples. The full-graph E1 loss combines participation BCE, candidate cross-entropy, and equal half-weights for own and counterfactual COMMIT supervision. During ten warmup epochs COMMIT uses teacher proposals. Afterwards exact labels are recomputed for the proposals actually queried; hard proposals separate gradients into decision modules. Training states remain teacher-generated. Rollout evaluation supplies the autonomous-state test.

Pedro's native comparisons instead use five seeds 0–4, common immutable pools of 800 cubic N40 graphs with three trajectories,17,760 updates and batches of 1,680 eligible root decisions. The direct GNN uses native-action losses. Its E1 follow-up uses 1,480 warmup updates and native own/counterfactual confirmation labels. Star-local and full-graph intervention samplers are different. The native follow-up also changes BE's auxiliary representation. Consequently its architecture comparison measures a training/representation package, not an isolated architectural intervention.

The integrated extension uses 24 new cubic N40 graphs, one trajectory each, eight validation graphs and two new seeds 73000–73001. Every teacher has the same five architecture cells. Timing pilots selected 500 common updates of 128 root decisions, with 50 warmup updates, before scientific pools were generated. This exposes 64,000 sampled root decisions per model. It is intentionally a short-budget study, not a repeat of 29,836,800-decision native training. All 70 final checkpoints are retained without validation promotion or a second training budget after observing outcomes. Compatible shared message/COMMIT initial weights are preserved across the three large palette models within a seed; the small family has different dimensions.

The two initialization replicas share teacher-specific data pools. They do not replicate data generation, and repeated rollout tapes do not create additional independently trained models. Hyperparameters, parameter counts, source hashes, training time, data manifests and checkpoint digests are recorded in `results/integration/multiteacher/`.

6.2 Evaluation and units of replication

The full-budget architecture cohort contains 304 common graph/tape cases per model, giving 1,520 cases for five seeds of one architecture. It covers degree and size transfer, irregular/heterogeneous families, cliques and structured controls. The native cohort contains 396 cases per model through N10,240. Its E1 follow-up uses already-published cases and is exploratory. Five seed outcomes on one graph are not five independent graph instances.

The new native extension uses five independent graph specifications: regular N16, irregular N40, regular N160, path 16 and K4, excluding training/validation labeled fingerprints. Every exact teacher and checkpoint produces 128 new independent executions on each graph. Full color vectors, statuses, native steps, failure labels and random seeds are archived. The graph ensemble is small and fixed; no confidence interval treats it as a representative sample of all graphs.

Participation, candidate and COMMIT accuracy use common teacher states and explicit private inputs. Native COMMIT labels retain their own semantics: ColorTrials preference rejection is not identical to a physical collision; GM assignment can deliberately create simultaneous conflicts before

repair. Per-phase action accuracy, component denominators and GM mark counts are retained so WAIT or negative-mark imbalance cannot alone establish learning.

Functional evaluation applies the model's own decisions, with native permissions but no oracle conflict veto. We report invalid outputs and incomplete outcomes alongside successes. Mean native steps over all attempts can be reduced by early failure; efficiency comparisons therefore also use successful outputs and remain within one teacher. Parameter count, training CPU time and native steps measure different costs. No deployment latency or cross-teacher LOCAL speed ranking is inferred.

6.3 Distributional inference

The primary observation is an independent complete run. For the historical local-list studies, V2 splits the valid completed prefix into selection and evaluation subsets, fits bounded witnesses on selection only, and calibrates their joint maximum with a full-vector MMD component on evaluation rows. Every permutation acts jointly on complete row labels; vertex columns are never independently shuffled. Pooled scaling ignores evaluation source labels. The permutation count 19,999 gives minimum p .00005.

The factorial primary family has 720 graph/checkpoint hypotheses and the robust study 795. Event, irregular-graph, historical-replication and explanation families remain separate. For the new extension, all 350 planned teacher/model/ graph comparisons form one Holm family. Every model is compared with its own teacher's exact reference. A cell with fewer than four valid completed rows is untestable and assigned p 1 only for planned-family bookkeeping. It is not an equality result. Conditioning, failure frequencies and the native cause of termination remain visible.

Pedro's separate behavior experiment uses a mixture of Hamming kernels and terminal-status equality, plus family/size-stratified permutation aggregation. It also tests a color-name quotient and obtains coupling-based TV upper bounds. We audit its exact statistic and compare procedures on matched rows, without pooling its hypothesis family with our graph-specific families.

7 Conditional correctness and representation results

Proposition 1 (safety and progress are separate). For a proper partial state, candidates supported on available colors and exact rejection of simultaneous equal neighboring offers preserve properness. This alone does not imply a uniform progress or termination bound.

Proof. A new color avoids old fixed neighbors by availability, and two adjacent equal proposals cannot both be confirmed. This proves safety by induction. In an all-free K-clique, suppose every node proposes color 0 and participates independently with probability 1/2. A node confirms only when it is the unique participant, so the probability that any node advances is $K/2^K$. Waiting for the first confirmation has mean $2^K/K$. A policy that always waits never terminates. Safety thus supplies no constant progress bound. The example concerns specified policies, not every learned checkpoint. □

Proposition 2 (exact teacher guarantees on the declared domain). Under fresh independent ideal private randomness and the hypotheses in Table 1, the uncapped LUBY, JOHANSSON and repeated ColorTrials teachers preserve properness and terminate almost surely, with O(log n) expected native rounds at degree$\leq$ 3. JP is safe and terminates within n uncapped native rounds. BE_BASE_D3 deterministically reaches four auxiliary colors in 15 native steps. Greedy

and synchronized speculative GM terminate under their supplied schedules and produce proper final colorings.

Proof sketch; complete arguments in AppendixA. Availability satisfies $|A_v| \geq d_U(v) + 1$. LUBY has conditional progress at least 1/4 using the independent candidate product and Jensen's inequality. For JOHANSSON each pending neighbor has at least two available colors, giving the sufficient degree-three bound 1/8. For ColorTrials, the event that a vertex has preference 3 for a free color and all at most three competitors have preference≤ 2 has probability at least 27/256. A uniform positive conditional progress boundρ gives $P(T > t) \leq n(1-\rho)^t$ by iterated conditioning and union bound. JP's eligible maxima form an independent set and include a maximum pending ID. Polynomial reductions preserve proper auxiliary colors; KW uses disjoint four-color blocks. Sequential first fit and the last-repaired endpoint argument complete the remaining cases. Finite caps are distinct from these uncapped guarantees. □

Proposition 3 (a finite-type sufficient COMMIT certificate). For the canonical MAX architecture, assume nonnegative deterministic shared messages, zero empty aggregation, legal candidate-slot gathering and pending/participation gating. A participating pending center's neighbor slot has six possible legal types. If the COMMIT logit has the correct strict sign on all 64 subsets of these types, it implements exact rejection at arbitrary degree and palette in real arithmetic.

Proof. Componentwise MAX is commutative, associative and idempotent, so only the set of present types matters, not multiplicities. The gather retains only the own candidate slot. Fixed ownership of that color and simultaneously participating equal offers are the two forbidden types. The 64-set sign table is therefore sufficient. It does not establish candidate diversity or termination. Native continuous-ID/context extensions are not automatically covered by this alphabet. Interval computations and floating-point error assumptions are documented separately. □

Proposition 4 (a scoped information limitation). A deterministic policy given only its own color/pending status and private draw cannot reproduce all arbitrary-state first-free decisions without neighbor information. Further, in the memoryless current-state interface, some random-trial COMMIT decisions require information beyond radius one.

Proof. A pending center with the same own inputs but a fixed neighbor of color 0 versus 1 has different first-free colors; its identical local input cannot prescribe both. For the radius example use path 0–1–2, with 0 and 1 pending and 2 fixed at 1 in one state and 2 in the other. Let both pending vertices participate, with candidate draws .3 at 0 and .4 at 1 in common K4. Vertex 0 proposes 1 in both states. Vertex 1 proposes 2 in the first state and 1 in the second, changing vertex 0's COMMIT. Its radius-one initial states/private inputs agree. This lower bound assumes no previously stored neighbor availability or supplied free-color flags. It is not a lower bound for every coloring algorithm, every persistent-memory protocol or every scheduling model. □

Proposition 5 (clique conditional-law equality). On K_n with exactly n colors, any vertex-equivariant protocol with exchangeable private input and positive probability of proper completion has the uniform conditional law on the n! proper labeled color vectors.

Proof. Equivariance and exchangeability give a vertex-invariant trajectory law by induction. The proper-completion event is invariant, so conditioning preserves this invariance. Vertex permutations act transitively on all proper n-color vectors. All conditional masses are therefore equal. Different candidate rules can satisfy the same conclusion. This is not proved by finite nonrejection and need not hold on other graphs. □

Proposition 6 (observable and coupling bounds). A simultaneous interval $[l, u]$ for $P_M(A) -$

$P_T(A)$ gives $\mathrm{TV}(P_M, P_T) \geq \max(0, l, -u)$. A valid coupling gives the upper bound $\mathrm{TV}(P_M, P_T) \leq P(X_M \neq X_T)$.

Proof. The first inequality follows by restricting the supremum defining TV to the eventA and propagating its confidence statement. For the second, the difference of probabilities of any event is bounded by the chance that the coupled outcomes disagree. Taking the supremum proves the bound. Neither compares TV distances across different-dimensional graph spaces. □

8 Learning across teachers: historical evidence

Table 2. Pedro's published full-budget native comparison. Each entry is successful graph/tape/checkpoint evaluations out of 1,980. These aggregate counts are preserved historical evidence; their original weights and raw output banks are missing here. The follow-up reuses known test cases.

Teacher	Direct-action GNN	E1 / declared native adaptation
LUBY	1,857	1,980
JOHANSSON	1,919	1,980
JP	1,635	1,631
COLOR_TRIALS	1,980	1,980
BE_BASE_D3	872	1,980
SPECULATIVE_GM	1,975	1,951
GREEDY	1,842	1,980

These results establish no universal teacher ranking. E1/adaptations improve the reported LUBY, Johansson, BE and greedy outcomes, leave ColorTrials at complete observed success, and do not improve JP or GM. BE's large change also changes auxiliary representation and training. Both positive and negative before/after comparisons matter.

The JP E1 follow-up has mean local action accuracy .999735 but succeeds in 1,631/1,980 autonomous evaluations. This directly illustrates accumulation and state-distribution effects; it does not identify a unique causal mechanism for every failure. The reported GM losses similarly warn that exact local validation on sampled teacher states need not guarantee all rare conflict marks in large autonomous executions.

9 Functional generalization and aggregation

Table 3. Full-budget historical architecture controls. Five independent training seeds share 1,520 held-out graph/tape/checkpoint evaluations per architecture; the fixed evaluation instances are blocked rather than treated as independent new graphs per seed.

Architecture / intervention	Reported principal outcome
E1 NORMALIZED	1,520/1,520 proper completions
DeepSets palette	1,520/1,520 proper completions
LocalPaletteTransformer	1,520/1,520 proper completions
Raw endpoint control	Lost progress in 346 evaluations

Architecture / intervention	Reported principal outcome
SUM aggregation	Stalled in 1,196 evaluations
MEAN aggregation	Produced 839 improper final colorings

The normalized/raw-endpoint intervention preserves scorer tensor shapes, initialization, data and update counts within seed, making it more controlled than the earlier raw-prefix architecture comparison. SUM and MEAN alter neighbor reductions and multiplicity handling. Their failures are evidence about those implementations and training conditions, not a proof that SUM or MEAN cannot represent any correct coloring policy.

The successful prefix-free models show that E1's original ordered prefixes are not necessary for observed functional generalization in this experiment. They still receive useful palette position information and learned graph messages. No conclusion says that generic attention alone supplies a distributed coloring algorithm. Constant parameter count also does not imply constant communication or inference cost.

10 Balanced teacher/architecture extension

Table 4. New balanced finite-cohort functional outcomes. Each model architecture/teacher cell pools two fixed initialization replicas, five fixed graphs and 128 independent runs per graph/checkpoint, totaling 1,280 attempts. The entries are counts of proper completions, not population confidence bounds or full-budget architecture rankings.

Teacher	E1	DeepSets	Transformer	Small	No messages
Luby	1,280	1,280	1,280	1,280	30
Johansson	1,280	1,280	1,280	0	0
JP	473	395	448	1	0
ColorTrials	1,280	1,280	1,280	640	0
BE (base d 3)	232	268	255	1	8
Speculative GM	1,156	1,170	1,155	444	0
Greedy	1,280	1,280	1,280	1,280	0

Figure 1 displays the same finite-cohort percentages. Individual seed/graph records, failure types and native steps are available in `functional.csv` and the original hashed banks. The exact reference uses one 128-run bank per teacher/graph shared across its model comparisons; it is not counted twice to create extra reference data.

All three large palette architectures completed 1,280/1,280 attempts for each of LUBY, JOHANSSON, COLOR_TRIALS, GREEDY. At the same short budget, JP and BE retained substantial failures, while speculative GM was intermediate. These differences compare specified training/representation packages, not intrinsic teacher complexity. Per-teacher architecture differences and both replicas remain in the table; there is no universal winner.

The short budget and small training graph sample limit interpretation. A failed cell can reflect optimization or coverage, and does not prove a representational impossibility. A successful small model establishes finite sufficiency within its family; it does not prove parameter minimality. The

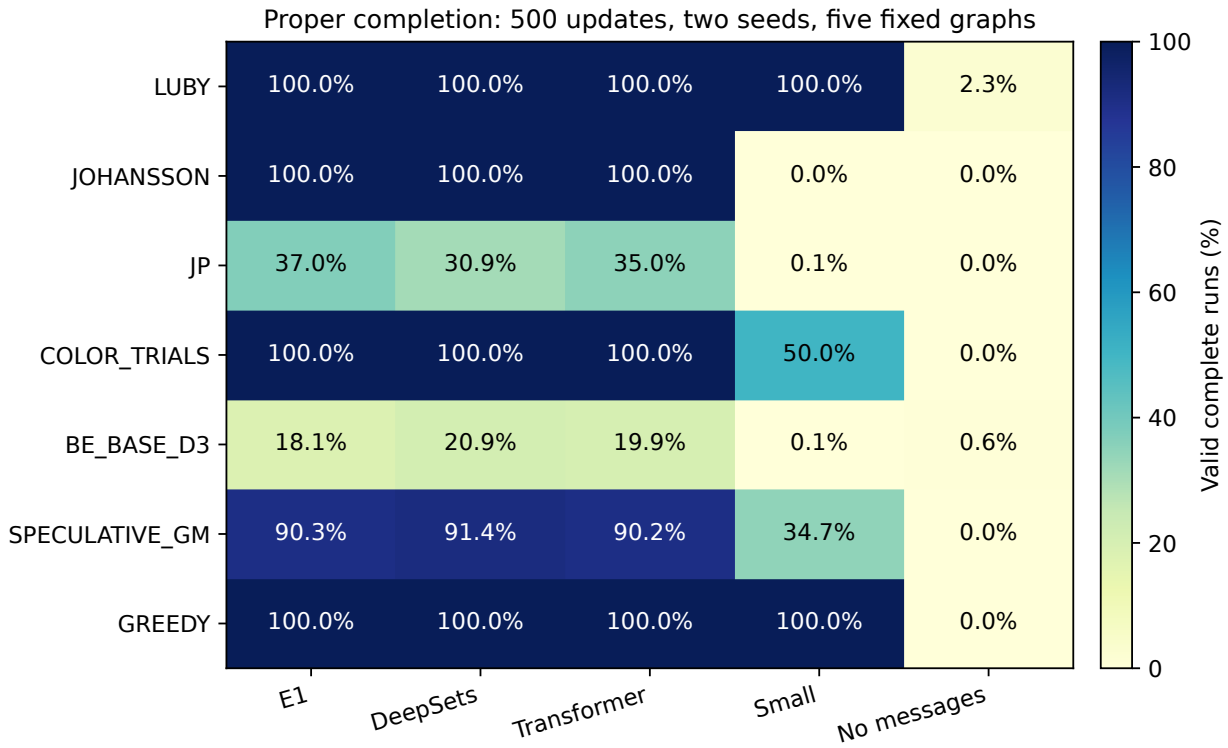


Figure 1: Proper-completion percentages on the balanced short-budget cohort. Every architecture receives 500 updates; each cell retains all failures and both training seeds.

no-communication control retains native permissions, so its successes cannot be generalized to a scheduler-free anonymous setting.

Figure 2 compares common-state local validation accuracy and autonomous functional validity for all 70 separately trained models. Local validation contains different phase/class mixtures across teachers, so between-teacher horizontal comparisons require the per-phase denominators. Each plotted point remains one checkpoint; rollout repeats are not training replicas.

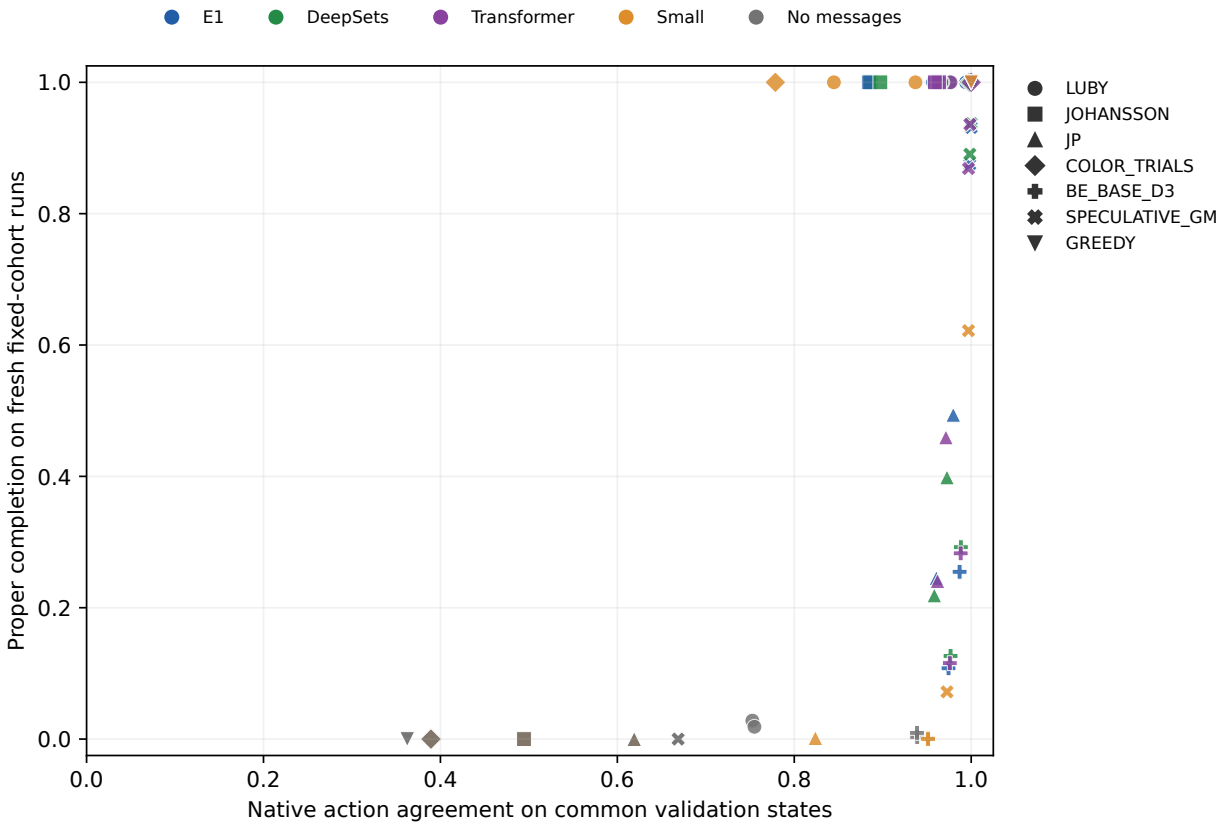


Figure 2: Common-state native action accuracy and autonomous proper-completion rate. Colors denote architectures and symbols teachers; all 70 checkpoints are plotted and identical points can overlap.

11 Complete-coloring laws and statistical reconciliation

11.1 Historical local-palette evidence

Table 5. Separate historical global-law cohorts. Counts are rejections after each declared Holm family. They are decisions under different cohorts, sample sizes and methods, not comparable numerical distances.

Architecture	Initial MMD:250 runs,265 comparisons	Robust V2:500 attempts,265 comparisons	Factorial V2:500 attempts,240 comparisons
E1	10	52	115
NORMALIZED DeepSets	46	107	170
Local palette Transformer	127	181	180

On identical new robust banks, the earlier single-scale MMD gives 21,71,147 rejections respectively; on identical factorial banks it gives 49,121,174. These matched-data comparisons isolate the procedure/budget difference from historical new-bank variation. They still do not turn rejection counts into a ranking of TV distances or make V2 uniformly more powerful. Archived auxiliary cases include losses as well as gains for V2.

The distributional differences coexist with strong functional generalization in the full-budget architecture cohort. Functional success restricts outputs to proper colorings but does not select a unique probability law on that set. The teacher itself need not sample proper colorings uniformly. Conversely, the clique symmetry argument gives genuine scoped equality despite different proposal mechanisms. Thus neither candidate mismatch nor success alone determines a global-law conclusion.

11.2 Reconciliation with the independent E1 behavior study

Pedro reports no rejections among 100 primary stratum/model/representation tests using five newly trained E1s, degree$\leq$ 3, common K4, cutoff 256 and 256 runs per graph. His experiment tests full labeled terminal vectors/status and a color-name quotient. It is not the same checkpoint cohort, degree/palette regime, cutoff or hypothesis family as Table 5. Ten labeled-law and eleven quotient-law pairs obtain simultaneous TV$\leq$ .05 certificates using coupling disagreement upper bounds. Those certificates are affirmative fixed-graph approximate-equivalence statements; zero rejections alone would not supply them. Their underlying raw banks and trained weights are not available here.

We independently verify the mixture statistic's equality to unbiased MMD and its inclusive permutation calculation. On identical archival 500-row banks from the first regular N128/degree 3 graph, neither mixture MMD nor V2 rejects any of the three s 41000 architecture comparisons. On the first regular N512/degree 100 graph, both reject all three. Each high-degree raw p-value is .00005; the exploratory six-test Holm value is .0003, whereas the original 720-test V2 adjusted value remains .036. The original statistics and p-values replay exactly. These are matched historical reanalyses, not new independent confirmation or a replay of Pedro's missing weights.

Different finite-sample power remains scientifically important. On a star with 200 iid binary leaves, change only the center from uniform on 199 high colors to uniform on a 100-color subset. Every vector is a proper local-list coloring, TV equals 99/199, and the mixture population MMD^2 is 9.569688e-6. Twenty fresh independent experiments per cell give Table 6.

Table 6. Independent matched-data controls, nominal alpha .05,999 permutations. Entries are detections out of 20 independent experiments.

Law	Runs per generator	Mixture MMD	Honest V2
Same-law star	256	0	1
Same-law star	500	1	1
Center-only alternative	256	0	20
Center-only alternative	500	3	20

For the alternative the pointwise exact 95% power intervals are[0,.1684] and[.0321,.3789] for the mixture and[.8316,1] for V2 at both sizes. Figure 3 preserves the null controls and uncertainty. This finite-law result does not establish strict Holm-adjusted power for 720 hypotheses or explain every historical model comparison. In particular, it does not show that Pedro's degree-three neural laws exhibit this star alternative.

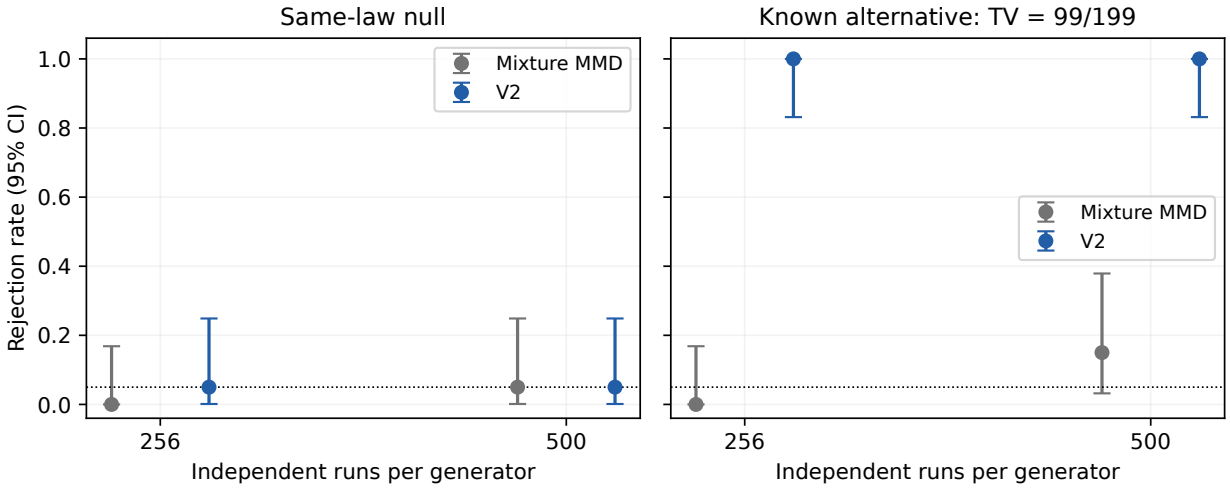


Figure 3: Independent known-law null and power checks. Both procedures receive identical fresh full-coloring rows in each trial; intervals are pointwise exact 95% binomial intervals over 20 experiments.

The discrepancy classification is consequently mixed: different output representations are different definitions; graph/palette/checkpoint/cutoff changes are protocol differences; the known-law power gap is statistical; and the effect of V2 on Pedro's exact original banks remains unresolved. No speculative explanation closes that final gap. The complete audit is `docs/integration/GLOBAL_DISTRIBUTION_RECONCILIATION.md`.

11.3 New teacher-specific global comparisons

Of 350 planned teacher/checkpoint/graph hypotheses, 227 are testable and 123 have fewer than four successful model/reference rows. After Holm over the complete planned family, 6 reject. Each reference bank belongs to the corresponding teacher. These counts concern conditional output laws on a small fixed cohort; functional failures and limited power remain separate.

All six new rejections concern two small checkpoints that nevertheless complete all 640 of their functional evaluations: LUBY seed 73001 and ColorTrials seed 73000. Each differs from its own teacher on three fixed graphs, with raw p .00005 and Holm-adjusted p .0175. The LUBY core has 551 parameters and the context-extended ColorTrials model 1,560. This supplies new finite-cohort

evidence for functional/distributional separation under two teachers. The other small ColorTrials initialization fails all 640 functional attempts, so the successful checkpoint is not presented as universal small-model robustness. The large architectures have no new rejections here; those nonrejections do not certify their law equality.

The native comparisons use the audited V2 selection/permutation construction with an explicit global-K4 histogram alphabet. The historical feature allocator assumed max(degree)+1 bins, which is insufficient for valid four-color native path outputs. The first call failed before producing a p-value. A separate adapter repairs that allocation while preserving actual degrees, feature definitions and the original frozen module. An independent color 3 path-null test and a matching-alphabet exact replay verify the compatibility change. This is documented as an integration repair, not an invalidation of the historical local-list study. The new native power is not inherited from its old validation by assertion.

Table 7. New Holm rejections / testable comparisons (ten planned per teacher and architecture).

Teacher	E1	DeepSets	Transformer	Small	No messages
Luby	0/10	0/10	0/10	3/10	0/2
Johansson	0/10	0/10	0/10	0/0	0/0
JP	0/8	0/7	0/7	0/0	0/0
ColorTrials	0/10	0/10	0/10	3/5	0/0
BE (base d 3)	0/6	0/7	0/5	0/0	0/1
Speculative GM	0/10	0/10	0/10	0/9	0/0
Greedy	0/10	0/10	0/10	0/10	0/0

Every law test conditions on proper native completion and compares with the corresponding teacher. The retained status/failure data are essential: a rarely successful network may have an estimable conditional law on some graphs while performing poorly overall. Untestable cells and nonrejections remain distinct. No test result establishes exact equality of all teacher and learner laws.

12 How size and degree affect distributional generalization

The factorial study varies n$\in$ {128,256,512,1024} and degree Delta$\in$ {3,10,50,100}, with three fresh regular graphs per cell and five fixed checkpoints per architecture. It uses local palettes,500 primary attempts and 125 separate explanation attempts per generator. Its target is this fixed graph/checkpoint cohort, not a population of infinitely many graphs or trainings.

Two preregistered events are $A = \{U(X) \leq \Delta\}$, with U the number of used colors, and $T =$ {color Δ occurs}. EventA says that at least one palette color is absent, not necessarily the maximum. For E1 at degree 100, the equal-weight cohort differences in P(A), model minus teacher, are approximately 0,.157,.639,.603 as n increases through the four values. Their simultaneous intervals include[.570,.703] at n 512 and[.550,.629] at n 1024. At degree 50, the top-color difference instead attenuates from-.239 at n 128 to-.004 at n 1024. These are observable-probability effects, not a monotone full-TV scaling law.

Figure 4 displays the registered effects with their simultaneous bands for all three architectures at degree 100. Across the principal factorial, the E1 eventA interaction between n 256/n 512 and

degrees 50/100 is .566 with simultaneous interval[.428,.704]. A shared teacher bank is counted once when averaging architecture effects. The earlier all-fifteen E1 replication at the new n 512/degree 100 graphs rejects 15/15 comparisons; combining the two prespecified event lower bounds gives a minimum simultaneous TV lower bound .078 across those fifteen graph/checkpoint pairs.

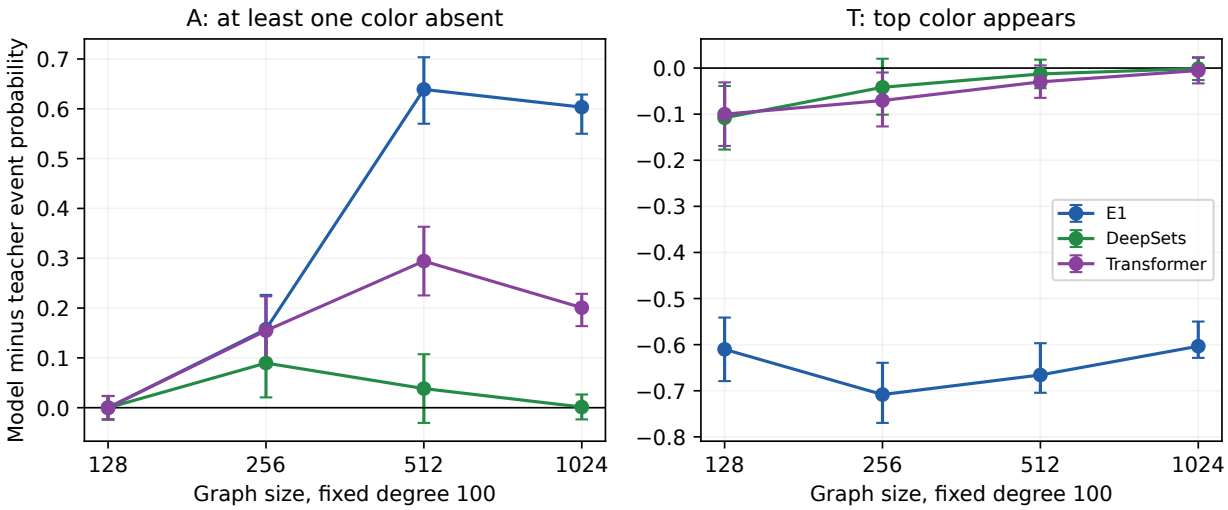


Figure 4: Fixed-cohort event-probability differences at degree 100, with simultaneous historical CP/Hoeffding bands. Curves concern two specific observables and do not order complete-law TV across graph sizes.

In regular graphs, degree also changes palette size. Their separate causal effects are not identified by this design. Comparing TV lower bounds on different graph spaces does not order the true TV values. Three graph instances and five historical trainings constrain population claims. The independent holdout witnesses and exploratory candidate interventions remain separate analyses; an observed proposal/final-law association is not alone a causal mediation result.

13 What the baselines establish about GNNs

Neighborhood information has a precise role: free-color decisions inspect fixed neighbors, and confirmation can require neighbors' availability-dependent offers. The scoped indistinguishability examples prove limits for an own-state or one-round current-state interface, rather than proving that a large GNN is necessary for graph coloring. Exact classical rules already solve the task with no trainable parameters when their required information and schedule are supplied.

MAX provides a degree-insensitive representation of Boolean existence and permits the finite-type COMMIT reduction. SUM can encode counts and MEAN proportions; a training procedure may exploit multiplicity-dependent shortcuts that fail outside the observed degrees. The historical negative controls demonstrate such risks for specific learned models. They do not constitute expressivity impossibility results for those aggregators.

DeepSets and attention investigate how a node represents its palette. Their success shows that the original prefix scorer is not the only sufficient mechanism in the tested setting. Both still communicate over graph edges. The small control tests reduced width within a normalized family. The no-communication control tests missing information while preserving core parameter count

and supplied permissions. Any comparison of those baselines must retain these differences.

The small LUBY model has 551 trainable parameters and completes all 1,280 LUBY evaluations; the small greedy control also completes all 1,280 under its supplied schedule. Thus the larger core is not empirically necessary for those finite cells. In contrast, the no-communication control completes 38/8,960 total attempts across the seven teachers. This supports the value of received information under this protocol, without proving that the particular large GNN or a learned message MLP is necessary. Small-model failures for other teachers are retained and do not establish incapacity beyond this budget.

Teacher independence is similarly scoped. One architecture can expose the action alphabet for all seven teachers, but supplied preferences, priorities, phases and schedules change the observation interface. Success across those interfaces is not learning arbitrary unseen algorithms from identical raw inputs. The finite experiment supplies evidence of adaptability, with cells that fail retained as limitations.

14 How Much Supervision Does It Take to Learn a Distributed Algorithm?

A teacher observation is one acquired decision label for a specified node, state and action component. In ordinary color phases we count participation, proposal and confirmation separately; Linial point selection and speculative conflict marking each count as one. Replaying a stored label adds optimizer work and no new supervision. Query counts, component types, trajectory clusters, unique observations and data hashes are recorded independently of updates. This distinction matters because sequential imitation changes the states a policy visits [Ross and Bagnell, 2010, Ross et al., 2011]. Existing imitation sample-complexity theory depends on its own state-space and horizon assumptions [Rajaraman et al., 2020]; our finite curves imply no analogous theoretical bound.

Information compatibility is established before comparing budgets. Anonymous regime A contains LUBY, JOHANSSON and repeated COLOR_TRIALS. Extended regime B contains JP with explicit relative-ID comparisons, BE with auxiliary colors, phase clock and supplied finite-field arithmetic, GM with priority comparisons and its native block/repair schedule, and sequential GREEDY with writable-node permissions. The three palette architectures have identical teacher-specific observations and learned graph messages. Transformer attention acts within a node's palette; it is not a replacement for graph communication. Interface sufficiency arguments and an anonymous-JP indistinguishability example do not prove universal representability by these fixed network widths.

For a policy to PASS, every frozen execution must end with a proper in-palette coloring, all protocol invariants respected, termination by the teacher-specific deadline, and no technical error, cutoff or incorrect terminal state. We use no 95% functional threshold, repair filter or failed-tape replacement. A teacher/architecture/budget combination passes only when all five independently trained policies pass their respective 1,000-case final batteries. These are five separate policies, not one policy tested 5,000 times.

All seven exact teachers passed every frozen final case before student confirmation. Native deadlines distinguish n-step sequential greedy and JP's at-most-n priority steps from BE's 15 reductions and the randomized teachers' prospective tail deadlines. Native steps with different semantics are never directly interpreted as identical distributed communication units.

The balanced cohort comprises 735 fresh trainings, five seeds per cell, seven query budgets from

1,000 to 1,000,000, and 500 Adam updates per training. Its shared teacher libraries acquired 34,999,967 component labels in total; smaller-budget policies only access their hashed prefixes. Training sizes are 8–24, development sizes 28/36, and confirmation sizes 32/48/64, in seven prescribed graph strata. The first budget where all five development policies are perfect nominates one frozen candidate; final results cannot choose another budget. Of 21 teacher/architecture packages, 16 nominated a candidate, 12 achieved five-seed PASS, and 80,000 final student executions were retained.

Regime A. Smallest validated candidate budget, or explicit lack of certification.

Teacher	E1: B*	DeepSets: B*	Transformer: B*
LUBY	1,000; 5/5 PASS	3,000; 5/5 PASS	1,000; 5/5 PASS
JOHANSSON	1,000; 5/5 PASS	1,000; 5/5 PASS	1,000; 5/5 PASS
COLOR_TRIALS	NOT CERTIFIED; 3/5 (candidate 300,000)	NOT CERTIFIED; 0/5 confirmed (no candidate; max 1,000,000)	NOT CERTIFIED; 0/5 confirmed (no candidate; max 1,000,000)

Regime B. Smallest validated candidate budget, or explicit lack of certification.

Teacher	E1: B*	DeepSets: B*	Transformer: B*
JP	1,000; 5/5 PASS	1,000; 5/5 PASS	1,000; 5/5 PASS
BE_BASE_D3	NOT CERTIFIED; 0/5 confirmed (no candidate; max 1,000,000)	NOT CERTIFIED; 0/5 confirmed (no candidate; max 1,000,000)	NOT CERTIFIED; 0/5 confirmed (no candidate; max 1,000,000)
SPECULATIVE_GM	NOT CERTIFIED; 4/5 (candidate 30,000)	NOT CERTIFIED; 4/5 (candidate 30,000)	NOT CERTIFIED; 4/5 (candidate 30,000)
GREEDY	1,000; 5/5 PASS	1,000; 5/5 PASS	1,000; 5/5 PASS

A NOT CERTIFIED cell reports an unachieved protocol threshold B*>1,000,000, not an impossibility theorem. Its earlier development budgets were evaluated; policies failing development were not exhaustively tested in confirmation. Per-seed queries, failures, compute and the distinction between an unattempted candidate and a failed confirmation remain in the machine-readable tables.

The grid does not assume monotonic improvement. Fixed optimization also means larger libraries can leave more acquired observations unused; visit counts are exported. Thus these are supervision curves conditional on the specified optimizer budget, not pure information-theoretic sample requirements.

For zero failures in 1,000 independent runs of one frozen policy, the one-sided 95% binomial lower success bound is 0.9970087504549047. Under fixed independent strata the same zero-failure bound conservatively controls their allocation-weighted mean by the arithmetic–geometric mean inequality. It does not cover every graph or simultaneously all trained policies. The five seeds remain the units of training replication.

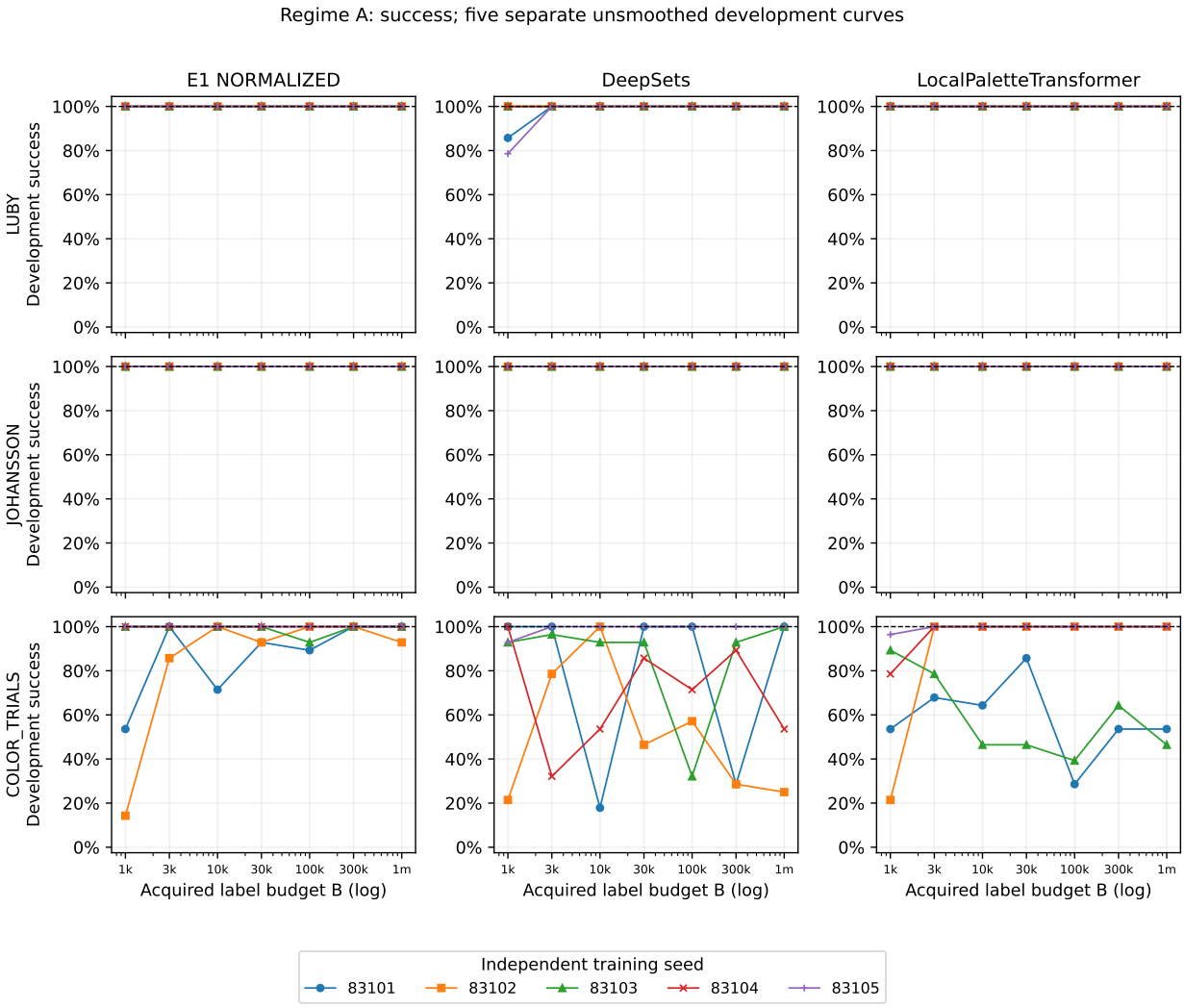


Figure 5: Anonymous-interface supervision curves. Each line is one training seed; these are development screens, not repeated final confirmations. The dashed line is the required 100% frontier.

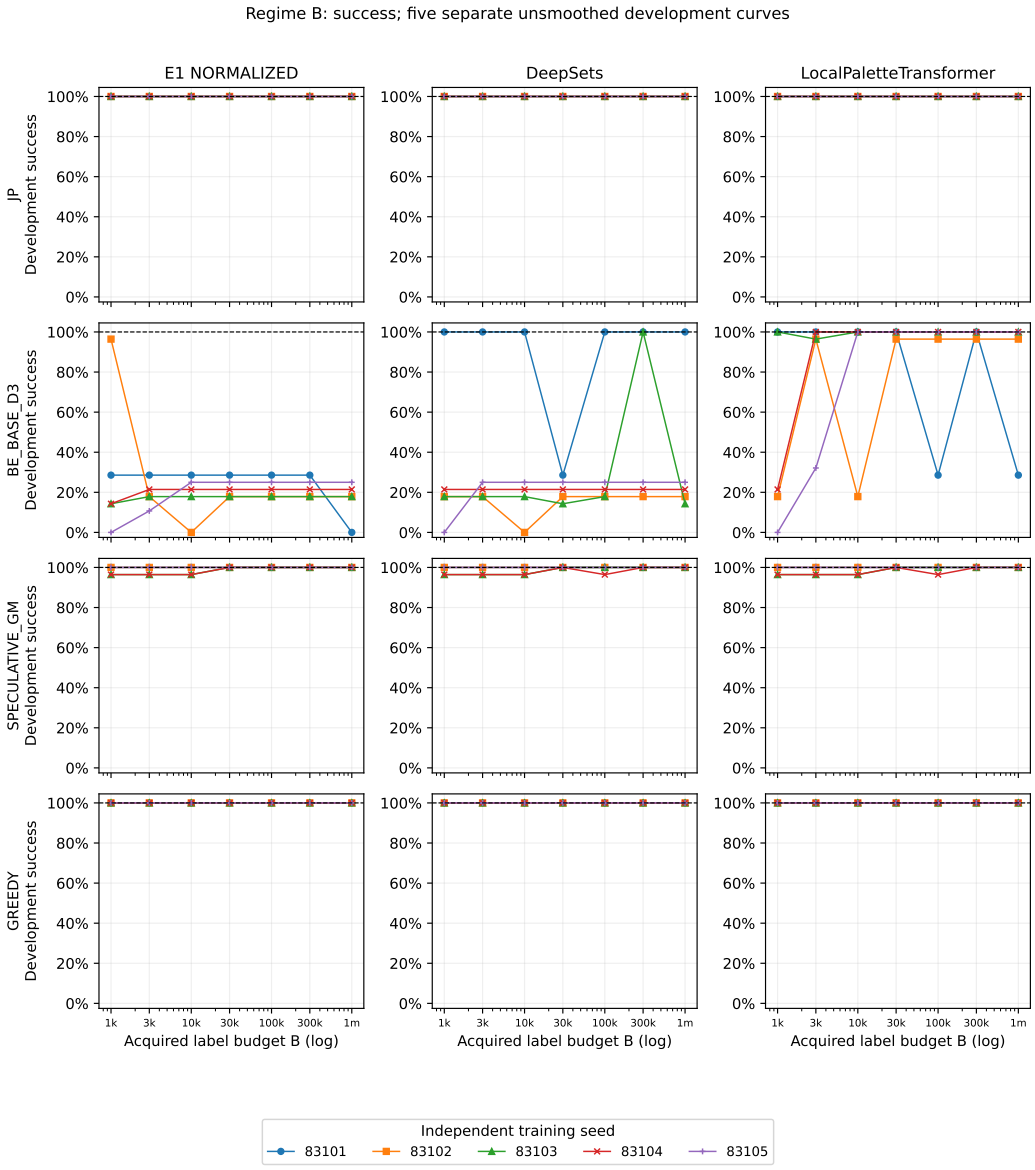


Figure 6: Extended-interface supervision curves with the same five-seed and 100% rules. Declines are retained without smoothing.

14.1 Optimization, capacity and interactive controls

The following entries count seeds completing all development cases, at the same B=3,000. They are exploratory development outcomes, never final PASS:

Teacher	Passive E1 500 updates	Passive E1 1,500 updates	Small 500	No messages 500	Interactive 500 total
LUBY	5/5	5/5	1/5	0/5	4/5
JP	5/5	5/5	0/5	0/5	5/5
BE (D3)	0/5	5/5	0/5	0/5	1/5

The fixed-label control changes only E1's optimizer updates. The small and no-message models share the same label prefix and update budget. The one-aggregation interactive control allocates half the labels to passive warmup and half to admissible states visited by the warm policy, with 250+250 updates and an Adam reset at aggregation. Failed learner trajectories remain recorded. It excludes GM's unproved arbitrary-history interface. These limited controls cannot identify every failure as a representation or data limitation.

Functional PASS does not logically imply action or transition fidelity, teacher-like execution time, or equality of complete coloring laws. Separate teacher-state fidelity and native-step curves retain all five seeds. Historical law tests are not imported as evidence for the new checkpoints. The full protocol, audit, negative results, dataset/checkpoint hashes and reproducible figure sources accompany this finite empirical statement.

The GM confirmation failures (normalized: 79, deepsets: 79, transformer: 79) leave only isolated vertices uncolored and occur in training seed(s) 83105. A post hoc singleton witness using the frozen candidate weights reproduces an incorrect confirmation rejecting assignment despite positive participation, while the exact teacher succeeds. The cached prefix contains 5 isolated-assignment bundles (three component labels per bundle), and the exact recorded minibatch order presents them 9 times; the affected seed's development suite contains 0 isolated vertices. This identifies a rare-state fit defect missed by development coverage. It does not show an absence of ID information or prove a universal capacity limitation. The diagnostic changes no model, budget, primary classification or final tape. Full traces and presentation-count checks are in structural-failure-witnesses.json.

The independent exploratory coloring-law supplement tests 210 fixed teacher/checkpoint/graph hypotheses using the audited native K4 robust instrument, split selection/evaluation rows, 9,999 permutations and Holm over the complete family. 210 comparisons are testable and 0 reject. Untestable comparisons retain p=1; conservative futility also cannot create rejection. These are conditional-on-success laws on two fixed small graphs, not equality guarantees or an additional functional PASS condition.

The small-model supplement repeats all seven budgets for three representative teachers and freezes independent final tapes under the same five-seed gate:

Teacher	Small parameters	Small validated B*	Seeds confirmed	Final attempted
LUBY	551	NOT CERTIFIED	0/5	False

Teacher	Small parameters	Small validated B*	Seeds confirmed	Final attempted
JP	1560	NOT CERTIFIED	0/5	False
BE_BASE_D3	1560	NOT CERTIFIED	0/5	False

Without a common development candidate no student final is attempted. These outcomes establish neither universal minimality nor the behavior of all possible small networks.

The additional fixed-label optimization protocol freezes both existing optimizer arms for all three control teachers and evaluates them on new paired final batteries. It checks identical label-prefix and initialization hashes and changes no primary outcome:

Teacher	500 updates, B=3,000	1,500 updates, B=3,000
LUBY	PASS (5/5)	PASS (5/5)
JP	PASS (5/5)	PASS (5/5)
BE_BASE_D3	NOT CERTIFIED (0/5)	PASS (5/5)

A PASS here validates the fixed B=3,000/optimizer point only; it is not a new grid minimum or a retry of the primary confirmation.

15 Frozen transfer across degree and palettes

The native coverage statement remains 21/21 under heterogeneous historical protocols, including five checkpoints and 1000 final runs per checkpoint. It does not mean that all policies used the same supervision or budget. We now extend the twelve passive sample-complexity packages and use the certified counterfactual checkpoints as the primary cohort for the nine new packages. Native matched/passive checkpoints remain secondary controls. No weights are modified or retrained in the transfer study.

Teacher and information audit. The exact repeated Luby, always-active Johansson, repeated ColorTrials, synchronous JP, four-block speculative GM and scheduled Greedy rules admit degree/palette parameterization. Both global lists of size $\Delta + 1$ and local lists $\{0, \dots, d(v)\}$ are justified. We preserve coherent ID orders, the four GM blocks, detection and serial repair, and Greedy permissions. The algebraic BE_BASE_D3 construction is excluded: its fixed fields and four-option reductions do not justify increasing the degree. A different Linial/KW or recursive BE variant would change the learning problem.

The historical rooted serializer fixes four colors and three neighbor positions. The three neural cores instead share weights across colors and neighbors. An explicit sparse structural adapter evaluates those same weights with variable palettes and the same raw predicates, signed IDs, phases and private inputs. It supplies no exact availability, winning flag, COMMIT or repair rule. Palette masks describe declared lists only. The two graph exchanges remain MAX; DeepSets and attention act within each palette. Across the 195 original selected/control checkpoints, 1170 new native executions succeed. For the 150 extensible checkpoints, 21630 teacher-state decisions match the

original wrapper, with no hard-action or final-color differences in these controls; the largest observed absolute logit discrepancy is 1.526×10^{-5}. These are finite parity checks.

Conditional exact-rule guarantees. With a proper partial coloring, $|A_v| \geq d_U(v) + 1$ under either list regime. Availability and exclusion of equal simultaneous offers prove random-trial safety. For ideal independent uniform proposals, Luby has conditional progress at least $1/4$. The implemented always-active Johansson variant also admits a self-contained bound: pending neighbors have at least two free colors, hence $q_u(c) \leq 1/2$; using $\log(1-x) \geq -2\log(2)x$ and Jensen yields $|A_v|^{-1} \sum_c \prod_u (1 - q_u(c)) \geq 2^{-2d_U(v)/|A_v|} \geq 1/4$. Discrete ColorTrials with preferences in $\{0, \ldots, \Delta\}$ has progress at least $(\Delta+1)^{-1}(\Delta/(\Delta+1))^{\Delta}$ for $\Delta > 0$. JP completes in at most n steps, Greedy in n, and GM in at most $2\lceil n/\min(4,n)\rceil + n$ scheduled steps. The random-tail bounds concern the ideal exact rules, not arbitrary learned proposals, a formal PRNG proof, or universal learned termination.

Frozen confirmatory experiment. After an exploratory degree/size pilot, we freeze graph/tape namespaces, checkpoint hashes, horizons, families and sample sizes before final outcomes. We validate all 150,000 exact-reference runs first. Graphs have maximum degree exactly requested. The 90 primary extensible policies receive 1000 runs at each of five conditions: global $n64/\Delta 3, 4, 10$, local $n64/\Delta 4$, and global $n256/\Delta 4$. Sixty certified matched/passive controls and thirty pre-existing CT controls at the same 4000-update point receive the four $n64$ conditions. The extra controls are not called historically certified. All runs use frozen weights; an additional CPU executor changes only scheduling, not the frozen design.

There are 810,000 confirmatory student executions, of which 808,413 are correct across distinct policies. The primary cohort contributes 449,791/450,000, including 90,000/90,000 on the common degree-three control. These totals are accounting, not a pooled success probability of one policy. Table 13 reports each teacher/core across all five training replicas. The pointwise two-sided 95% Clopper–Pearson interval for 1000/1000 is approximately $[.996318, 1]$, under the independent uniform-family mixture model; no simultaneous or universal claim follows.

Sixteen of eighteen extensible teacher/core combinations pass every primary condition in all five replicas. Johansson/Transformer seed83101 obtains 972/1000 at global degree ten, with 28 clique cutoffs. GM/Transformer seed94102 obtains 819/1000 on local degree-four lists; its global conditions pass. E1 and DeepSets pass all selected primary conditions for all six extensible teachers. Other Transformer replicas and conditions are retained individually. Successful transfer therefore depends on both palette regime and checkpoint, not only a single maximum-degree number.

The exploratory pilot spans $\Delta = 2, 3, 4, 5, 10, 20, 50, 100, 200$ at regular $n256$, and sizes 64–1024 at fixed degree3 or10, with additional irregular, hub, tree, bipartite, disconnected, isolated and clique cases. It gives 3526/3570 correct outcomes. The remainder are two matched CT collisions and 42 dense CT resource cutoffs. Some degree50 runs completed before a pilot resource amendment; unfinished dense CT cells then received a ten-second cap. Those outcomes do not have a homogeneous resource horizon. All five teachers other than CT have successful degree200 pilot cases in each replica, but one run per checkpoint/cell does not certify that range.

A separate post-confirmation exploratory size extension gives 240/240 correct executions at regular $n2048/4096$, degrees3 and10, for all sixty primary Luby/Johansson/ColorTrials/JP checkpoints. It uses an independent namespace and one run per checkpoint/point, after validating eighty exact references. It is not included in the confirmatory denominator or a size-range certificate.

Teacher	Core	$\Delta 4$ global	$\Delta 10$ global	$\Delta 4$ local	$n256, \Delta 4$
ColorTrials	DeepSets	5/5	5/5	5/5	5/5
ColorTrials	E1	5/5	5/5	5/5	5/5
ColorTrials	Transformer	5/5	5/5	5/5	5/5
Greedy	DeepSets	5/5	5/5	5/5	5/5
Greedy	E1	5/5	5/5	5/5	5/5
Greedy	Transformer	5/5	5/5	5/5	5/5
Johansson	DeepSets	5/5	5/5	5/5	5/5
Johansson	E1	5/5	5/5	5/5	5/5
Johansson	Transformer	5/5	4/5 (972–1000)	5/5	5/5
JP	DeepSets	5/5	5/5	5/5	5/5
JP	E1	5/5	5/5	5/5	5/5
JP	Transformer	5/5	5/5	5/5	5/5
Luby	DeepSets	5/5	5/5	5/5	5/5
Luby	E1	5/5	5/5	5/5	5/5
Luby	Transformer	5/5	5/5	5/5	5/5
GM	DeepSets	5/5	5/5	5/5	5/5
GM	E1	5/5	5/5	5/5	5/5
GM	Transformer	5/5	5/5	4/5 (819–1000)	5/5

Table 13: Frozen primary policies: checkpoints with 1000/1000 successes out of five. Nonperfect cells additionally give the per-checkpoint success range out of 1000. The first three columns use $n = 64$; the last uses the global palette at $n = 256$. Each condition samples the declared uniform mixture of ten exact-degree families. There is no averaged policy.

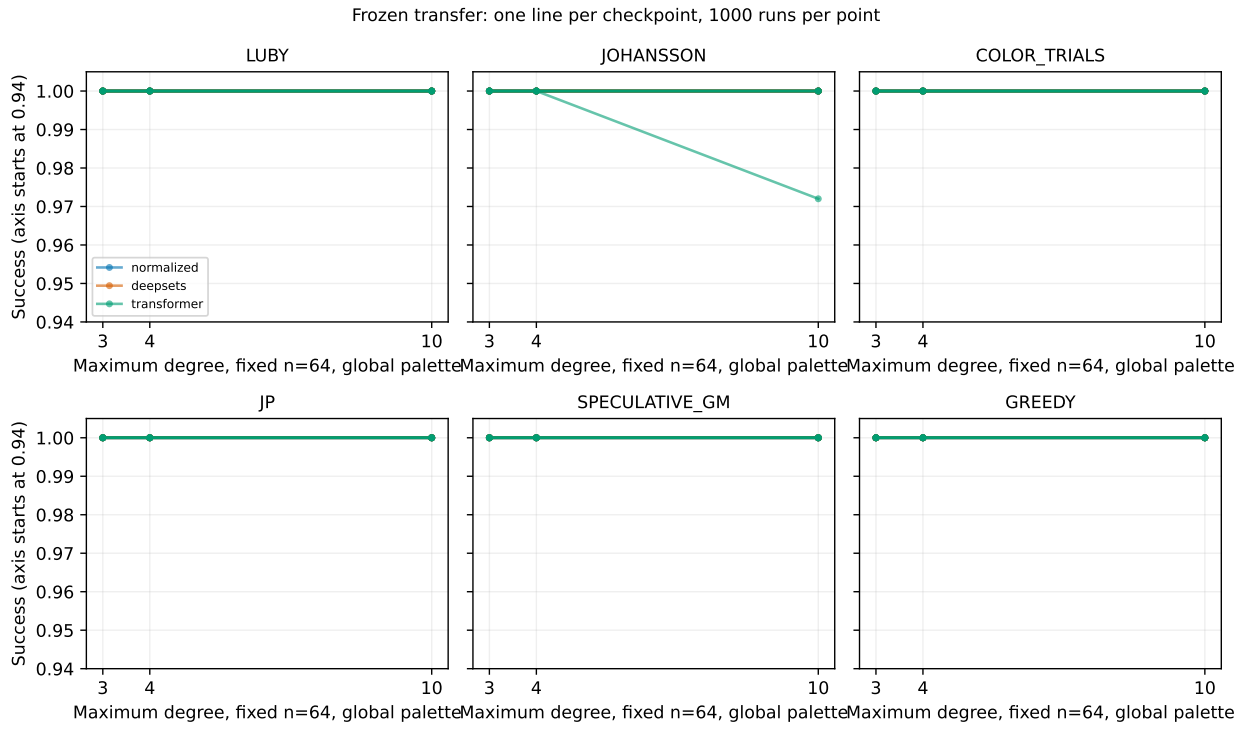


Figure 7: Frozen degree transfer at fixed $n = 64$ and global palettes. Each line is one training checkpoint, with 1000 runs per point. The success axis starts at .94 to expose rare failures. Curves do not average the five policies and do not separate degree from palette size.

Why functionality does not establish the learned rule. A typed CT preference intervention isolates a local limitation. Place an edge in one component and a degree-Δ star in another. The edge endpoints always have degree1 and local lists $\{0, 1\}$. Give the first endpoint preference Δ for color0, the second $\Delta - 1$, and zero otherwise. The first is the strict exact winner. The conservative input normalization makes the gap $1/\Delta$. All fifteen primary CT CF policies accept the queried winner at degrees3–5 but reject it from degree10 in this probe. Neighbor count and receiver palette stay fixed; the wrong decision is not caused by aggregating more neighbors. These are valid typed private-input counterfactuals, not a claim that a recorded PCG64 seed emits the entire constructed vector, nor a random-rollout failure estimate.

The primary encoder uses preference divided by global degree, agreeing with the native preference/3 template. A separate posthoc observation encoding keeps the old preference/3 units at all degrees. On this same raw-gap1 query it accepts 13/15 winners at degree5, 5/15 at10, and0/15 at50 and200, versus15/15,0/15,0/15 under the primary encoder. Both preserve preference order; neither resolves every measured case. The main frozen results are not replaced by an outcome-selected normalization. These witnesses establish sensitivity to the input scale/range, not information loss or impossibility under every possible embedding.

The Johansson failure exposes another mechanism. A full-batch replay reproduces 128 steps and the original cutoff. Two adjacent pending vertices remain with available colors $\{0, 2\}$; both propose color2 in the final rounds and COMMIT correctly vetoes the collision. In a separate conditional probe, each vertex proposes color2 for all 8192 new private draws. This observed concentration does not prove exactly zero mass elsewhere, but explains the lost progress in the recorded trace. Figure 8 separates this proposal-diversity failure from CT's comparison failure.

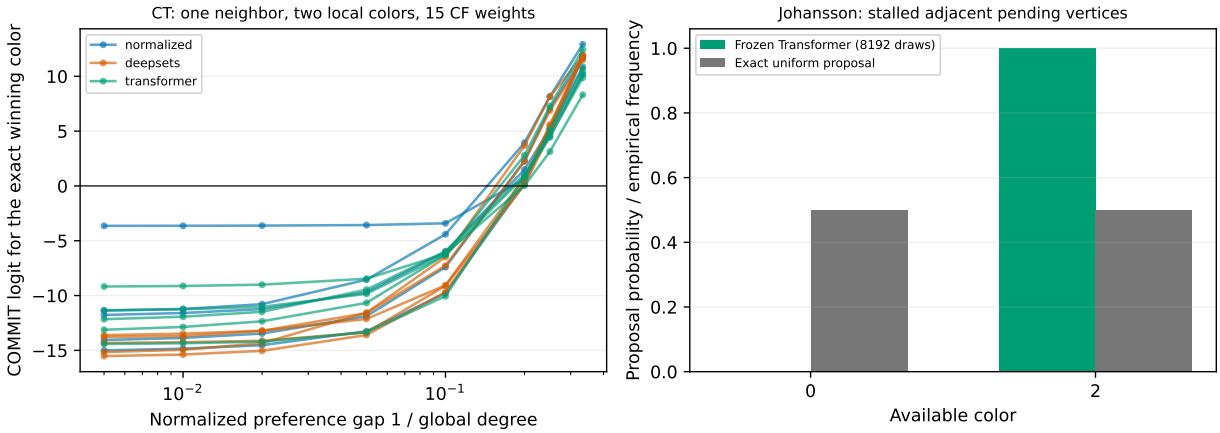


Figure 8: Left: fifteen CT counterfactual checkpoints queried on the same winning color as the normalized preference gap shrinks, with one neighbor and two local colors. Right: one stalled Johansson/Transformer state; the two pending receivers have identical observed histograms, shown for one of them, versus the exact uniform proposal. These are component diagnostics, not complete-coloring distribution tests.

For GM, the same bad local-list repair proposal occurs in CF, matched and passive seed94102. CF rejects that proposal while the other heads accept it, but the external schedule does not revisit the missed repair; all three retain a conflict in the diagnostic case. Thus improved conditional COMMIT response need not improve final functionality. Learned alpha values are latent features,

not directly supervised calibrated availability probabilities; threshold disagreement is not itself a decision error.

A degree-three control separates the GM palette effect. On an edge plus a disconnected three-star and isolates, coherent IDs place both edge endpoints in the same speculative slot. Exact prefixes require the same repair color1 under global K4 and local $\{0, 1\}$. All three Transformer seed94102 procedures choose color1 globally but color0 locally; the other four seeds choose correctly. Thus this failure can occur without increasing degree: it is a local-palette transfer limitation, not evidence that degree4 alone causes the error. These constructed states are diagnostic, not extra finals.

Exploratory COMMIT probes use 120 policies and 838,188 typed queries on exact safe prefixes. At matched initializations, native minibatches and update counts (CT4000, GM500), CF improves mean per-seed query fidelity against matched, with effects depending on degree and architecture. CF also uses more labels and changes the loss distribution. These controls therefore do not isolate counterfactual supervision at equal total label budget, nor prove a monotone functional benefit in degree. Full paired denominators and all raw head errors are retained.

Finally, the historical outward-interval test of 64 finite MAX message supports certifies neither exact COMMIT nor absence of false acceptance for any of the 45 selected passive canonical Luby/Johansson/Greedy cores. This concerns conditional head inputs, not necessarily states reachable under their own proposals or Greedy's single-writer schedule. It does not contradict their finite functional success, and cannot certify progress. The interval/float32 arithmetic assumptions are recorded separately.

The completed extension supports a bounded addition to this article. Multigrade retraining, degree/palette causal separation, label-budget-matched CF, uniform learned progress, and a high-degree BE variant remain distinct future experiments; none is run here.

16 Discussion

The evidence supports the proposed distinction among notions of learning. The native JP example separates local action accuracy from autonomous validity. The historical palette study separates successful coloring from exact candidate imitation. Full-law tests and event bounds separate task validity from teacher-distribution fidelity. Clique symmetry and scoped coupling certificates show why the converse statement, that every imitating network must have a different global law, would also be false.

Teacher changes can affect a policy without changing its unrestricted output law. Exact synchronous JP with fixed distinct priorities gives the same first-fit solution as sequential greedy in descending priority order. If ID order is uniform, ascending and descending orders are both uniform, so uncapped JP and random-order greedy induce the same final-coloring law. Their common-ID executions and native costs can differ. Conditioning JP on a restrictive 128-round cap can also change that equality. This elementary observation prevents equating algorithm identity with solution-law identity.

The statistical reconciliation illustrates a complementary point. A characteristic kernel identifies laws in the population but can place a large TV difference in a weak finite-sample direction. Honest local witness selection can help in a specified alternative class while preserving conditional calibration. It can still miss rare or unrepresented high-order effects. Statistical instruments are part of the experiment, not neutral or universally powerful labels attached afterwards.

The strongest paper narrative therefore concerns the relationship between teacher specification, architectural information and the notion of fidelity. It does not require deciding that one research line was right and the other wrong. Differences caused by definitions should be stated; actual bugs should be repaired with evidence; remaining numerical gaps should remain open. This approach permits a defensible joint interpretation without erasing contradictory-looking historical records.

17 Limitations and unresolved claims

The principal historical limitation is missing collaborator runtime data: five retrained E1s,35 native E1s, source pools and original full coloring banks. Published tables and sources can be checked, but exact model-level reproduction has not been completed. The new 70-model cohort is not a substitute for those artifacts. Independent scientific review of this integrated draft remains pending.

The historical balanced integration design has two initialization replicas,24 training graphs, one trajectory per graph and a much smaller exposure budget than the earlier full-budget studies. It does not identify an asymptotic architecture ranking or separate optimization from capacity in failed cells. The historical rooted serializer by itself does not support native teacher transfer above degree three or to local lists. The explicit frozen structural extension in Section 15 addresses six teachers on its stated checkpoints and finite conditions. Separate historical local-palette evidence retains its own checkpoints and protocols.

The separate supervision study uses five replicas and controlled acquired-label budgets but only 500 primary updates and a finite graph/tape distribution. Its additional fixed-label optimization and interactive controls probe selected failure mechanisms; they do not identify an intrinsic information-theoretic sample requirement or establish a universal architecture ranking. Its local prospective freeze is not an independently timestamped external registration.

Finite functional success does not prove universal neural safety, termination or complexity preservation. Canonical MAX certificates require precise algebraic/numerical assumptions and do not cover every native context head. The simulator is synchronous, with supplied schedules and arithmetic for some teachers; distributed preprocessing and asynchronous deployment are not evaluated.

Output-law tests have finite power and resolution. Their nonrejections are not equality statements, while conditional solution laws do not determine failure or stopping-time laws. Multiple separate historical families do not give one 5% guarantee over their union. CP/Hoeffding/DKW bounds concern their stated finite cohort or observable families, not a graph population. No causal degree/palette separation or general monotone TV law is established.

18 Counterfactual supervision across native teachers

18.1 Reconstruction and causal domains

We reconstruct M6B from authenticated original source and 21 historical artifacts. The 249/500 traditional, 476/500 own-proposal and 500/500 counterfactual outcomes concern one training seed, 10001, on paired cubic N10240 cases. Thirty original-policy executions are replayed exactly. M6B uses the 8967-parameter M4B architecture, whereas canonical E1 has 16035 parameters. The own/CF checkpoints share every participation, first-message and candidate tensor; only the second-message and COMMIT tensors differ. The configured 120-epoch horizon is 17760 updates; the selected checkpoints are epoch 53, or 7844 updates. The traditional comparison also changes the

number of tapes per graph, so its improvement is not a pure CF effect.

Pedro's 16 source hashes are verified against commits fc756cef, 80f486eb and 6836d855. None of the 40 expected original E1 weight hashes was recovered from 6614 local checkpoint files or the inspected public release manifests. His five 1980/1980 results, JP 1631/1980 and GM 1951/1980 remain authenticated reported evidence with five initializations over 396 published cases and a common training pool. His 17760 updates of 1680 roots and own/CF training also differ from the primary 500-update, batch 64 passive study. New runs below reproduce mechanisms, not those unavailable original policies.

We define a phase-specific label operation

$$\mathcal{O}_T(S, a, \xi) = D_T^{\phi}(R_T^{\phi}(S, a, \xi)),$$

where S is an admissible native snapshot, a a typed intervention, ξ the private input, and R recomputes every causal descendant before the phase decision D is queried. We distinguish native decisions from defensive continuation queries on offers that the native policy might not emit. The latter are additional supervised tasks, not a claim that arbitrary simulator arrays are reachable teacher states. For BE and JP, forced participation cannot bypass phase eligibility or priority; ColorTrials confirmation requires strict winning preferences under actual availability. Linial points and GM detection marks have separate native oracles.

Admissible contexts are reconstructed from legal prefixes. Changing IDs requires a new complete replay, including comparisons, blocks, permissions and auxiliary initial colors. Private preference/draw interventions preserve the persistent state and regenerate messages. KW arithmetic and Linial evaluations are recomputed; speculative GM histories retain their supplied schedule, actual assignment colors, marks and repair queue. We export 162 additional hard contexts with replay witnesses. The three palette cores share these training-only adapters. All teacher and autonomous inference source files remain unchanged from the baseline; no exact decision replaces the neural confirmation head.

18.2 Paired controls and strict new certifications

We train 135 primary runs: three pending teachers, three cores, five new seeds and three treatments. Each preserves 500/1500/4000 checkpoints, the same native bank, initialization and minibatch sequence within a seed/core. Passive is the historical loss. Matched evaluates COMMIT with its stored teacher proposals and corresponding label. Fixed-bank CF uses 100 matched warmup updates, then P + C + .5 matched confirmation + .5 CF confirmation. Each normal root acquires three native component labels and four additional stored confirmation queries; Linial and DETECT each acquire one native label. Own-proposal queries in a separate seven-teacher pilot are charged individually.

Training uses N8/12/16/24. An amendment recorded before the additional screen and all new finals expands development from 60 to 300 cases at N28/36. The first common perfect checkpoint is frozen per treatment. Final has five independent 1000-case batteries at N32/48/64, ten fixed family strata, native horizons, and the strict original invariant/terminal gate.

Matched passes all nine pending packages, using 1500 updates for BE, 500 for ColorTrials/E1, 1500 for its other cores and 500 for GM. CF initially fails BE at 1500 in seed 94105 for all cores, and ColorTrials/E1 at 500 in seed 94103. Independently frozen follow-ups compare existing 1500/4000

CF checkpoints on new namespaces, retaining both outcomes. BE 1500 again fails in seed 94105; ColorTrials/Transformer1500 has two failures in its new battery. All six 4000-update packages pass. The following CF fixed points therefore close all nine cells, while the historical twelve remain separate: coverage is 21/21, not one homogeneous newly retrained cohort.

Teacher	Core	Updates	Native labels	All labels	Final
BE	normalized	4000	3000	6776	5 x 1000/1000
BE	deepsets	4000	3000	6776	5 x 1000/1000
BE	transformer	4000	3000	6776	5 x 1000/1000
ColorTrials	normalized	4000	30000	70000	5 x 1000/1000
ColorTrials	deepsets	4000	30000	70000	5 x 1000/1000
ColorTrials	transformer	4000	30000	70000	5 x 1000/1000
GM	normalized	500	3000	6104–6120	5 x 1000/1000
GM	deepsets	500	3000	6104–6120	5 x 1000/1000
GM	transformer	500	3000	6104–6120	5 x 1000/1000

The complete extension retains 195000 final execution rows, 194979 correct, and 170 distinct frozen checkpoints. Repeated checkpoints on a second battery are not new training replicas. All eight representative failed checkpoint/battery traces reproduce their original outcomes under the same 32-case cohort and 1024-root numerical batching. BE failures are omitted required KW recolorings that violate the new auxiliary bound. No failure is repaired or removed. An explicit erratum preserves and corrects the CT follow-up plan's reason-text typo (the original E1 failure was at 500, not1500); the actual checkpoint hashes, batteries and rows are unchanged.

18.3 Functional validity and continuation fidelity

At 4000, all permitted roots in 60 common native development trajectories per seed are audited, with four fresh continuation queries per color root. The table reports descriptive aggregate percentages across five policies; root decisions are not independent training replicas. Per core, the CF denominators are 499200 for BE, 41036 for GM and 57576 for ColorTrials. Full per-seed action, actual-transition, component and phase counts remain machine-readable. The certified GM500 policies have separate full audits at their actual checkpoint; the table fixes4000 for the paired comparison. CF improves the broader continuation task, while the matched/passive procedures already satisfy the functional gate. Thus CF is not uniformly necessary, or better at the same update count, for native functional success. A matched ColorTrials/E1 run regresses from 60/60 to 43/60 development successes between 1500 and4000 updates.

Teacher	Core	Matched CF (%)	CF-trained CF (%)	CF native actions (%)
BE	normalized	9.269	99.993	100.000
BE	deepsets	9.269	99.993	100.000
BE	transformer	9.269	99.993	100.000
GM	normalized	75.363	100.000	100.000
GM	deepsets	75.363	100.000	100.000
GM	transformer	75.363	100.000	99.995
ColorTrials	normalized	97.256	99.745	99.187

Teacher	Core	Matched CF (%)	CF-trained CF (%)	CF native actions (%)
ColorTrials	deepsets	99.097	100.000	100.000
ColorTrials	transformer	98.076	100.000	100.000

For BE/GM, every second-stage and COMMIT tensor is identical across the three palette cores in each paired fixed-proposal seed/update comparison. Those losses do not use the candidate scorer, whereas ColorTrials reads its candidate after the second exchange and couples those gradients. Shared heads explain correlated architecture outcomes; they do not provide three independent confirmations of a head-level effect.

Nested ColorTrials banks at native budgets 1000/3000/10000/30000 and fixed 500/1500 updates provide development-only supervision curves. Every CF label is charged; these do not establish a new certified minimum. A matched GM control without the added isolate families also passes all five new development seeds, so the tested ablation does not isolate a positive coverage effect. The authenticated old seed 83105 failures and all 30 new regressions remain a separate observation. The own-plus-CF portability pilot passes 60/60 development in six teachers for one new E1 seed, but BE obtains 22/60 at 1500; copying the Luby curriculum is insufficient evidence of a uniformly successful multiteacher procedure.

18.4 Information, complete outputs and transfer

JP's information control fixes the architecture, labels, batches, 500 updates and initialization, changing only two local context channels. Relative order passes all five 60-case development suites; raw endpoint IDs, differences and removed IDs do not. Twenty numerical replays reproduce every original ablation weight tensor and retain intermediate Adam/RNG checkpoints.

JP representation	Development successes, five seeds (/60)
without_ids	6, 6, 6, 6, 6
numeric_ids	13, 11, 6, 26, 15
numeric_differences	16, 15, 19, 31, 20
relative_comparisons	60, 60, 60, 60, 60

An indistinguishability witness swaps the IDs on a pending edge: anonymous observations are equal but native eligibility reverses. Edge order suffices for the implemented priority decision. This is an observation-level obstruction, not a claim that numeric IDs are unlearnable. ColorTrials has a second-hop availability witness. Historical M6 also aliases the float64 draws $1/2 - 2^{-28}$ and $1/2 + 2^{-28}$ to the same float32 observation, despite opposite activation labels. The current native teacher directly queries its supplied float32 inputs, avoiding that particular obstruction.

After functional selection, all 45 successful CF policies are frozen before fresh law and transfer tapes. On path8 and K4 plus four isolates, 256 shared-input runs per graph/policy compare the full labelled terminal vector and status, without success filtering. The coupling inequality and exact one-sided binomial bounds with Bonferroni .05/90 yield 89/90 simultaneous TV-at-most-.05 certificates. These are approximate fixed-graph marginal-law bounds, not equality. ColorTrials/E1 seed 94104 on path8 has36/256 coupled disagreements, upper bound .223630 and mean 2.191406

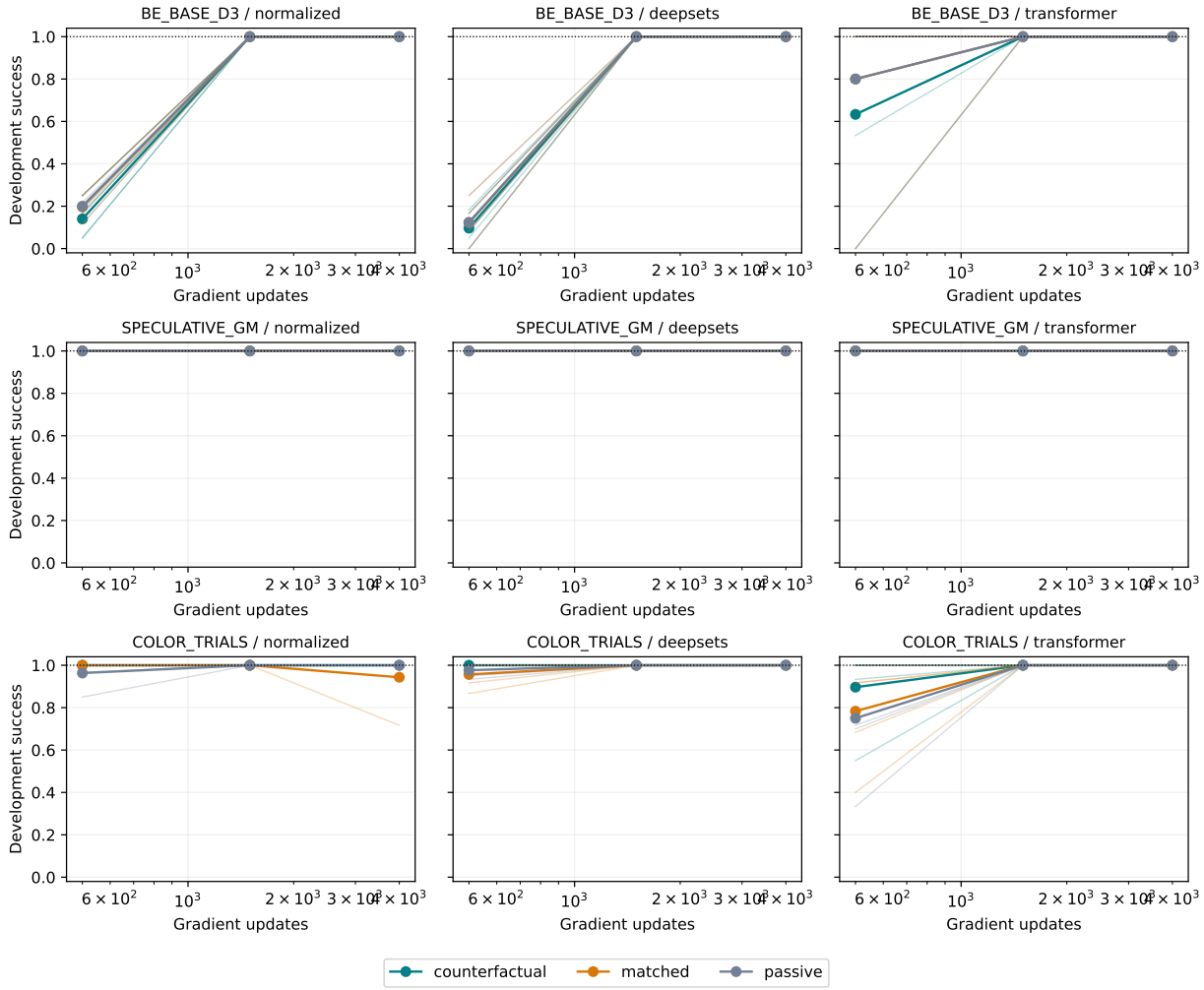


Figure 9: All five unsmoothed development curves and descriptive means at fixed native banks. Perfect development is a selection screen, not a functional certificate.

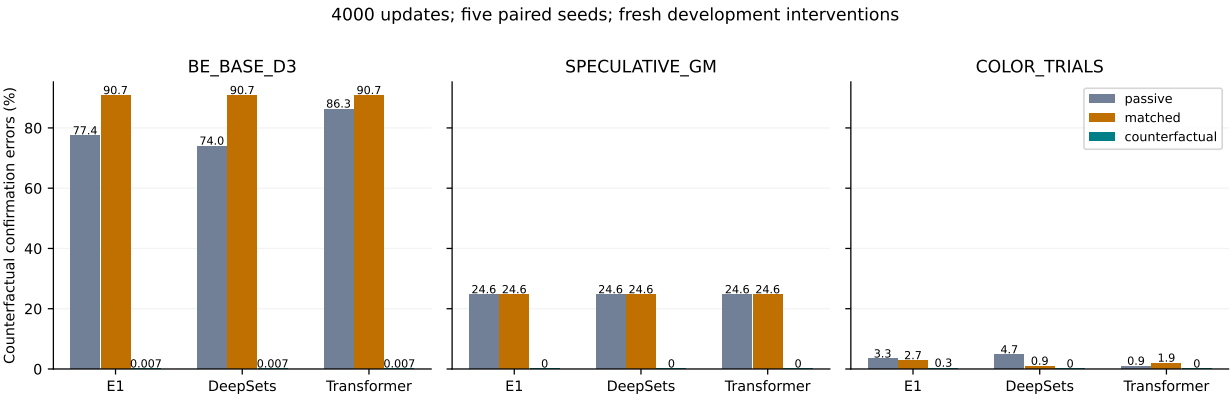


Figure 10: Errors on fresh continuation queries, separated from native action fidelity. Small nonzero errors retain explicit labels.

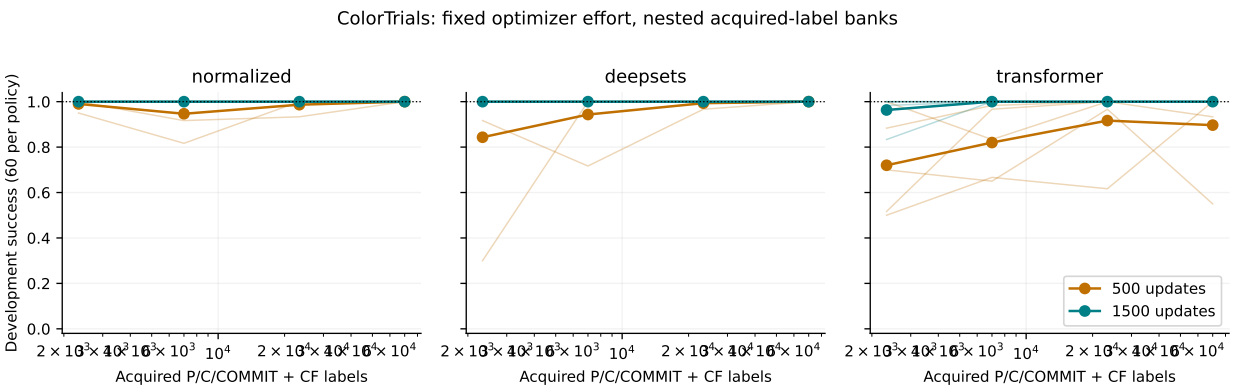


Figure 11: Supervision and optimization are distinct axes; thin lines show all five seeds. These lower-budget points have no new final certification.

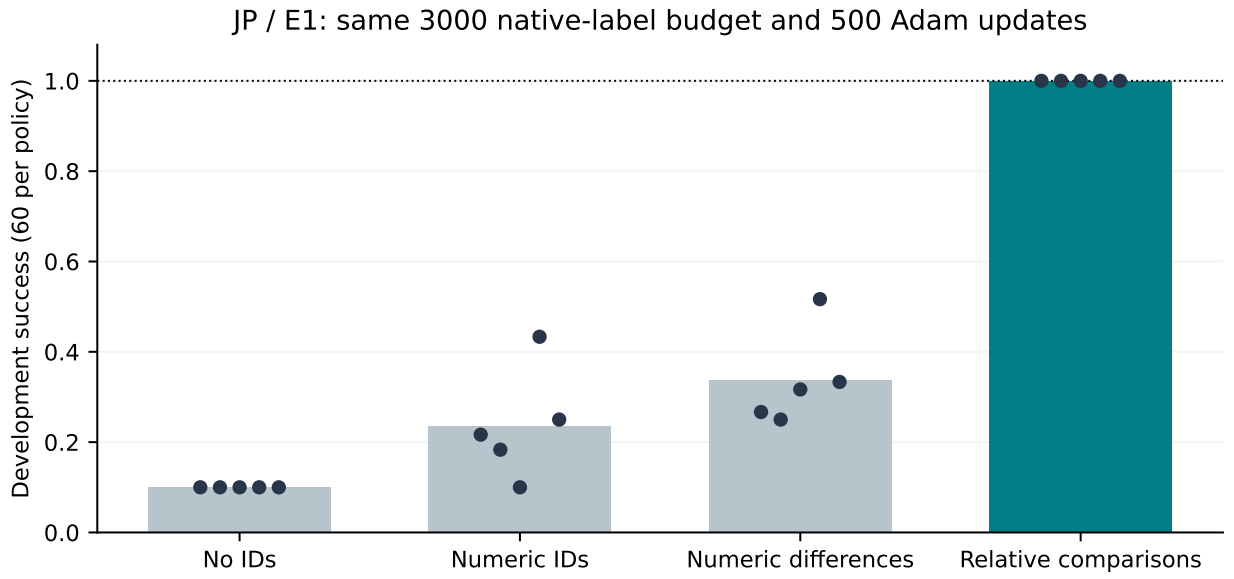


Figure 12: Information representation at a fixed optimizer and label point. Each dot is one training seed; development performance is not final certification.

native steps versus 2.097656. Coupled disagreement is not a TV lower bound or proof that the marginal laws differ. Empirical full-vector histograms and native times are retained separately.

Exploratory transfer obtains 3780/3780 correct executions, 84 per policy at N128/256/512. Its ten domain families are supplemented by binary trees, diamond chains, triangle chains and subdivided K4 components, all of maximum degree3. These finite observations do not extend the teacher to higher degree, local palettes or arbitrary graphs/tapes. Functional validity does not establish universal decision or law identity.

The complete campaign contains 278 computational training runs, including paired arms, pilots and numerical replays, with 683504 gradient updates and 2.580 CPU optimization hours. Training GPU time is zero; the original M6 CUDA replay is separate. Process times overlap. Acquired components, reused labels, presentations, simulator work and optimization remain separate; no sample-efficiency claim compares unequal update budgets. All sources, models, datasets, witnesses, negative results and manifests accompany this extension. The verification recomputes every final gate and all stored CF labels with zero label inconsistencies. The full suite's639 cases and29 later added cases pass.

These contributions complement trace-oriented CLRS Veličković et al. [2022] and learner-state acquisition in DAgger Ross et al. [2011]. Recent contrastive graph-coloring work Le et al. [2026] studies approximate-coloring objectives. Our contribution is the specified admissible-oracle framework, controlled separation of training factors, and finite native execution evidence; no comprehensive bibliographic priority or universal learnability claim follows from this study.

19 Conclusions

Learning from an algorithmic teacher has several distinct targets. Our integrated coloring project shows how the teacher's information and action interface constrain architecture design, how strong local imitation can fail in autonomous execution, and how functional extrapolation can coexist with a changed complete-solution law. Exact teacher baselines and smaller controls prevent attributing classical task solvability to GNN complexity alone. Symmetry and coupling also identify settings where distributional agreement is possible despite different decision mechanisms.

The counterfactual extension supplies independently validated functional points for all nine pending packages, preserving the prior twelve as historical reference. Its controls show that label concordance and optimization can close a functional gap without demonstrating universal CF superiority. Counterfactual continuations, shared-head dependencies, information representation and complete-output laws remain separate scientific objects, with every failed point retained.

These conclusions are conditional on the stated protocols and evidence. The repository preserves all cohorts, negative results, repaired failures, checkpoint identities and reproducibility boundaries. Completing historical collaborator artifact recovery and independent scientific review is necessary before presenting the entire joint project as independently reproduced or submission-ready.

20 Appendix A: teacher proofs and execution audit

For permanent trials, at most d(v)-d_U(v) distinct colors are fixed nearby, so both a four-color list at degree$\leq$ 3 and a local d(v)+1 list satisfy $|A_v| \geq d_U(v) + 1$. Conditional independence gives

$$P(J_v = 1 \mid q) = p_v \sum_{c \in A_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)).$$

For LUBY, $p_u = 1/2$ and the center law is uniform on A_v. The chord inequality $\log(1 - x) \geq 2x \log(1/2)$ on[0,1/2], followed by Jensen for the exponential, gives $P(J_v = 1 \mid q) \geq (1/2)(1/2)^{d_U(v)/|A_v|} \geq 1/4$. For JOHANSSON every pending neighboring candidate has mass at most 1/2 on any color, because its list contains at least two colors. Thus its collision-free product is at least $2^{-d_U(v)} \geq 1/8$ here. A ColorTrials vertex wins some color whenever its preference for one available color equals 3 and all eligible neighbors' preferences are$\leq$ 2, giving 27/256. Iteration of the conditional noncompletion bound and a vertex union bound prove almost-sure termination and logarithmic expected native rounds. These arguments refer to fresh independent mathematical random inputs; the documented finite-grid numerical simulation is a separate object.

For JP, distinct adjacent local maxima are impossible and a largest pending ID always exists. Availability and induction prove safety; at least one pending vertex is fixed each round, giving at most n uncapped rounds. The path 129 increasing-ID counterexample proves that 128 is not a universal fixed termination cap. To see equivalence with descending-order first fit, induct along the larger-neighbor dependency DAG. Every vertex sees the same already-colored higher-priority neighbors in both executions. Uniform ID order makes its reversal uniform too, proving the unrestricted output-law identity with ascending random-order greedy. A24-permutation path 4 test checks the finite example, without replacing the argument.

For Linial initialization, encode auxiliary colors below B as base-p polynomials of degree$\leq$ r. Distinct adjacent auxiliaries give nonzero polynomial differences; each has at most r roots in the prime field. With p>3r, some point avoids all at most three neighboring equalities. Adjacent nodes choosing the same output pair would have equal values at the same point, a contradiction. The implemented fields 13 and 7 satisfy these inequalities for B65,536 and 169, producing 49 auxiliary colors.

For KW set $q = \lceil 4B/5 \rceil$. Retain auxiliaries below q. Give each old color $a \geq q$ the disjoint block $\{4(B-1-a), \ldots, 4(B-1-a)+3\}$. All blocks lie below q because $4(B-q) \leq q$. At most three neighbor colors forbid choices, so a free choice exists. Recoloring endpoints have disjoint blocks, and recolored nodes avoid retained neighbor colors. Properness is preserved while the bound decreases. From 49 it reaches 4 after 13 such steps, giving 15 with the two field steps.

During exact GM assignment, a later slot avoids all earlier assigned colors; only same-slot assignments can conflict. Detection marks the lower-ID endpoint of every such conflict, so the unmarked subgraph is proper. In sequential repair, four colors suffice and the last repaired endpoint of an edge avoids the other's final color. Thus the final coloring is proper. For maximum block width w and m marks, the supplied schedule uses 2w+m steps. Greedy similarly finishes after n first-fit steps. Neither proof turns a sequential scheduler into an anonymous LOCAL algorithm.

Independent scalar neighbor loops match 7,641 native transition choices over 103 unlabeled simple graphs of at most six vertices and degree$\leq$ 3, with two specified tapes per graph/teacher:1,442 complete executions with no observed failures. Eight explicit random-array comparisons additionally match canonical and native LUBY at every state. This is finite-domain evidence over the sampled IDs/tapes, not exhaustive universal verification. The missing simple-undirected input guard was reproduced by 21 failing cases and repaired without changing supported graph rules. Full traces,+pre/post regression files and proof assumptions are archived.

21 Appendix B: certificates and statistical guarantees

The canonical second-stage candidate-slot input has six legal neighbor types: fixed-other, fixed-same, pending/waiting/other-offer, pending/waiting/same-offer, pending/active/other-offer and pending/active/same-offer. Nonnegative shared messages and MAX reduce arbitrary multiplicities to 64 subsets. For an active pending center with one legal candidate slot, the own input is fixed. The fixed-same and pending/active/same-offer types exactly characterize rejection. Checking every strict sign therefore suffices in real arithmetic. Stored float 32 weights are interpreted as exact rational coefficients for interval analysis; floating-point error enclosures additionally require their stated IEEE rounding, overflow and precision assumptions. This does not certify arbitrary CUDA kernels or every context-extended teacher head.

For fixed n and gamma>0, the one-coordinate Hamming kernel matrix on any finite color alphabet is $(1-a)I + a11^T$, with $a = e^{-\gamma/n} \in (0,1)$. Its eigenvalues are strictly positive. The full-vector product matrix is a Kronecker product; restriction to proper colorings is a principal submatrix and stays positive definite. Hence distinct finite probability vectors have positive population MMD^2 . This proves identifiability, not good power for a given graph and m. A terminal-status delta gives separate positive-definite blocks; a quotient representation instead changes the observed space.

Conditional on independent witness-selection data, V2's evaluation statistic is fixed and the pooled scaling ignores source labels. Under equal conditional solution laws, independently sampled evaluation row labels are exchangeable given the pooled observations and group sizes. The inclusive rank of the observed statistic among uniform random whole-row permutations, including the original, yields the super-uniform plus-one p-value. Ties are conservative. The argument makes no independence assumption about vertices within a row. The declared-alphabet native histogram adapter preserves this proof because its features are deterministic and its selection split is unchanged.

For Holm, suppose m 0 hypotheses are true. If any true hypothesis is first rejected, its ordered p-value is at most alpha/m 0. A union bound on the m 0 super-uniform true-null p-values bounds this event by alpha, regardless of dependence between tests. This covers shared reference banks within each stated family, not an uncorrected union of multiple families.

The factorial event inference uses a saturated binomial probability per graph/generator bank. Architecture means give equal weights to five fixed checkpoints and three fixed graphs, retaining the shared teacher term once. Simultaneous Clopper–Pearson intervals receive total failure budget .025; weighted Hoeffding contrasts receive .025 across 1,800 prespecified contrasts. Their intersection has simultaneous coverage at least .95 by union bound under the stated independent-run fixed-bank model. These are event-risk contrasts, not confidence intervals for a population of graphs or a general TV scaling parameter.

22 Appendix C: evidence map and reproducibility boundaries

ID	Claim	Experiment	Evidence boundary
C01	Full-budget functional extrapolation	paper-study	Reported archive plus immutable artifact manifests; finite cohort only

ID	Claim	Experiment	Evidence boundary
C02	Native teacher comparison	Pedro native	Published aggregate evidence; original raw/weight reproduction incomplete
C03	Pedro E1 output-law findings	Pedro behavior	0/100 reported rejections; coupling certificates reported, raw verification pending
C04	Teacher correctness	Exact native rules	Conditional proofs plus 1,442 finite-domain runs; no universal learned-policy inference
C05	Matched full-law comparison	Six archived banks	Original V2 statistics replayed; exploratory historical reanalysis
C06	Known-law sensitivity gap	Fresh independent control	20 trials per law/size; pointwise CP; no universal power conclusion
C07	Balanced multiple-teacher learning	New 70-model cohort	Fresh banks, two initializations, short budget; fixed graphs, not population rankings
C08	New conditional global laws	New native K4 cohort	350 planned Holm comparisons; untestable cells distinct from equality
C09	Size/degree observable effects	Historical factorial	1,800 simultaneous fixed-cohort contrasts; degree and palette confounded
C10	Input guard repair	New regression	21 failures before, pass after; no supported historical result invalidated
C11	Declared alphabet repair	New integration compatibility	Failed before path p-value; exact matching-alphabet replay and null regression

ID	Claim	Experiment	Evidence boundary
C12	Architecture necessity limits	Specified observations	Scoped information arguments and finite sufficiency, no global minimality theorem

The repository's `docs/PAPER_SCIENTIFIC_AUDIT.md` supplies the substantive independent-review handoff, including claims that remain unsupported. Raw integration banks and all 70 final checkpoints accompany their SHA256 manifest. Historical recovery releases include the local-list banks and 85 selected paper-study checkpoints. The separately listed missing Pedro weights/banks prevent claiming full joint historical reproduction.

The earlier manuscript and historical result sections remain recoverable from preserved branches and the archived pre-integration manuscript. Code organization retains the `src/dcl/` package to preserve checkpoint and import compatibility. Plot/table generation is separate from training and sampling. Repeated analysis regenerates data-derived artifacts, not new favorable results. Exact commands and their tested scope are in the README.

Acknowledgments

The preserved Git history attributes the degree/palette and mechanism studies to Martín Ríos-Wilson and the native-teacher/retrained-E1 studies to Pedro Montealegre. This draft preserves that provenance without settling author order. Original algorithm authors and software dependencies retain their existing attributions. The repository does not invent an accepted publication or a license absent from the project.

References

Petar Veličković and Charles Blundell. Neural algorithmic reasoning, 2021. URL https://arxiv.org/abs/2105.02761.

Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. The CLRS algorithmic reasoning benchmark. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 22084–22102, 2022. URL https://proceedings.mlr.press/v162/velickovic22a.html.

Nathan Linial. Locality in distributed graph algorithms. *SIAM Journal on Computing*, 21(1): 193–201, 1992. URL https://courses.csail.mit.edu/6.852/08/papers/Linial.pdf.

Ryoma Sato, Makoto Yamada, and Hisashi Kashima. Approximation ratios of graph neural networks for combinatorial problems. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1905.10261.

Andreas Loukas. What graph neural networks cannot learn: Depth vs width. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/pdf?id=B1l2bp4YwS.

Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017. URL https://arxiv.org/abs/1706.03762.

Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 627–635, 2011. URL https://proceedings.mlr.press/v15/ross11a.html.

Olga Solodova, Nick Richardson, Deniz Oktay, and Ryan P. Adams. Graph neural networks gone hogwild. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2407.00494.

Michael Luby. Removing randomness in parallel computation without a processor penalty. *Journal of Computer and System Sciences*, 47(2):250–286, 1993. doi: 10.1016/0022-0000(93)90033-S. URL https://www.sciencedirect.com/science/article/pii/002200009390033S.

Öjvind Johansson. Simple distributed $\Delta+1$-coloring of graphs. *Information Processing Letters*, 70 (5):229–232, 1999. doi: 10.1016/S0020-0190(99)00064-2. URL https://www.sciencedirect.com/science/article/abs/pii/S0020019099000642.

Mark T. Jones and Paul E. Plassmann. A parallel graph coloring heuristic. *SIAM Journal on Scientific Computing*, 14(3):654–669, 1993. doi: 10.1137/0914041. URL https://doi.org/10.1137/0914041.

Fabian Kuhn and Roger Wattenhofer. On the complexity of distributed graph coloring. In *Proceedings of the ACM Symposium on Principles of Distributed Computing*, 2006. URL https://ac.informatik.uni-freiburg.de/publications/publications/podc06.pdf.

Leonid Barenboim and Michael Elkin. Distributed $(\Delta + 1)$-coloring in linear (in Δ) time, 2008. URL https://arxiv.org/pdf/0812.1379v2. Version 2; the implemented teacher uses only the degree-three base case.

Johannes Schneider and Roger Wattenhofer. A new technique for distributed symmetry breaking. In *Proceedings of the ACM Symposium on Principles of Distributed Computing*, 2010. URL https://tik-old.ee.ethz.ch/file/22a700b74fc2f37acb56d690e62c86cf/podcfp107_schneider_188.pdf.

Assefaw Hadish Gebremedhin and Fredrik Manne. Scalable parallel graph coloring algorithms. *Concurrency: Practice and Experience*, 12:1131–1146, 2000. URL https://eecs.wsu.edu/~assefaw/publications/Concurrency12-2000.pdf.

Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012. URL https://www.jmlr.org/papers/v13/gretton12a.html.

Jesse Hemerik and Jelle Goeman. Exact testing with random permutations. *TEST*, 27:811–825, 2018. doi: 10.1007/s11749-017-0571-1.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6:65–70, 1979. URL https://www.ime.usp.br/~abe/lista/pdf4R8xPVzCnX.pdf.

Stephane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 661–668, 2010. URL https://proceedings.mlr.press/v9/ross10a.html.

Nived Rajaraman, Lin Yang, Jiantao Jiao, and Kannan Ramchandran. Toward the fundamental limits of imitation learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/1e7875cf32d306989d80c14308f3a099-Abstract.html.

Thien Le, Tianyu Zhao, and Melanie Weber. Contrastive neural algorithmic reasoning for graph coloring, 2026. URL https://arxiv.org/abs/2606.03923.